

Machine-Learning-Assisted Empirical Bayes Estimation for Regression on Grouped Network Data

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Abstract

We study the latent group structure in a two-way effect model for matched data represented by a bipartite network. A directed acyclic graph (DAG) encodes the assumed relationships among the observed outcomes, network links, latent effects, group memberships, and prior parameters, thereby clarifying the data-generating process and facilitating estimation. Unobserved heterogeneity is modeled using group-specific prior distributions, allowing individual effects to vary both across and within groups. Although the two-way effects can be estimated by ordinary least squares (OLS), this estimator may be imprecise when the underlying bipartite network is sparse or connected across latent groups. We therefore develop a machine-learning-assisted EB estimator that first learns the latent group structure from the network and then exploits this structure to shrink the estimated effects toward their corresponding group-specific distributions. We establish identification conditions, consistency of prior-parameter estimates, oracle equivalence, and the conditional MSE/bias–variance characterization. The Monte Carlo simulations show that it substantially reduces estimation error relative to OLS under the data-generating processes considered. Finally, in an empirical application to Amazon user–product ratings, we use the proposed method to estimate and characterize heterogeneity in the user- and product-specific components of ratings.

Keywords: Two-way effect model, empirical Bayes method, directed acyclic graph, structural causal model, Bayesian network, community detection method, bipartite graph, matched data, shrinkage estimation.

JEL Codes: C11, C13, C18, C23, C55

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1 Introduction

Two-way fixed-effects regression has long been a workhorse for estimating high-dimensional heterogeneity in models based on interactions or matched data, dating back to the seminal contribution of (Abowd et al. (1999), AJPS). Important applications include the estimation of worker and firm effects (Card et al., 2013), student and teacher effects (Jackson et al., 2014), firm and bank effects in lending relationships (Amiti and Weinstein, 2018), patient and healthcare-location effects (Finkelstein et al., 2016), and family and neighborhood effects (Chetty and Hendren, 2018), among many others. More recently, the literature has connected conventional two-way effect models to network analysis by representing matched observations as a bipartite network. Within this framework, Jochmans and Weidner (2019) establish a direct connection between the statistical properties of least-squares estimators of two-way effects and the connectivity structure of the network induced by the matched data. In particular, weak network connectivity can substantially reduce the precision with which individual effects are estimated, generalizing the well-known limited-mobility problem in applications. Motivated by this limitation, Cheng et al. (2025) develop a shrinkage approach and propose an empirical-Bayes estimator that regularizes the high-dimensional effects using a prior distribution informed by assortative matching patterns in the observed network. By borrowing information across nodes in the bipartite network, their estimator can improve the estimation of individual effects when the network provides insufficient information for precise unrestricted least-squares estimation. The main goal of estimating a two-way effects model on a bipartite network is to recover and conduct inference on the node-specific effects on the two sides of the matched data, and especially to understand when those effects can be estimated accurately given the network’s connectivity structure.

This paper studies the latent group structure in a two-way effects model defined on a bipartite network induced by matched data. Matched data naturally induce a bipartite network in which the two sets of nodes correspond to the two types of observational units and links represent observed matches between them. For example, in a student–teacher setting, a link is present between student i and teacher j whenever student i takes a course taught by teacher j . Connecting all observed student–teacher pairs therefore generates a bipartite network in which links occur only across the two node types, rather than among students or among teachers. Jochmans and Weidner (2019) exploit this network representation of matched data to study two-way effects and show that estimation of the individual effects is intrinsically related to the connectivity

structure of the induced bipartite network. We extend this framework by allowing the bipartite network to exhibit a latent group structure on both sides. Such a structure is economically natural in many matching environments. For instance, students majoring in mathematics are more likely to take courses taught by mathematics instructors, while still taking some courses offered by other departments such as economics, computer science, or statistics. This generates relatively dense interactions between particular groups of students and instructors together with less frequent cross-group links. Similar patterns arise in worker–firm data, where positive assortative matching implies that more productive workers tend to be employed by more productive firms (Abowd et al., 1999; Kline, 2024). These examples suggest that observed matches may contain systematic group-level structure beyond individual heterogeneity and overall network sparsity. Our model explicitly incorporates such latent group structure into the bipartite-network representation of the two-way effect model, allowing matching patterns across groups to provide additional information about the heterogeneous effects on both sides of the network.

Our second contribution is to introduce a group-specific prior structure for the two sets of heterogeneous effects. Rather than imposing a single homogeneous prior distribution on all individual effects, we allow the distributions of the two-way heterogeneity to vary systematically across latent groups. For example, productive workers share a common prior distribution for their individual effects, while less productive workers are governed by a corresponding group-specific prior. The latent network structure therefore determines how information is pooled across observational units in the empirical-Bayes estimation, while still allowing substantial heterogeneity among individuals within the same group. This specification is motivated by the distributional evidence in Bonhomme et al. (2019), who show in the worker–firm setting that workers and firms can be organized into groups according to worker quality and firm productivity and that matching frequencies vary systematically across such groups. In particular, relatively high-type worker groups tend to interact more frequently with relatively high-type firm groups, and relatively low-type worker groups tend to match disproportionately with relatively low-type firm groups. We provide an explicit network representation of this idea by modeling the observed matches as a bipartite network with latent group structure and linking the estimated network memberships to group-specific prior distributions for the two-way effects. In this way, the proposed framework connects systematic group-level matching patterns in the network to the distributional heterogeneity of the underlying individual effects.

Our third contribution is to introduce a directed acyclic graph (DAG) as an integral component

of the estimation framework. DAGs have long been used in econometrics and statistics to clarify model structure, conditional dependence, and the assumptions underlying regression specifications (Imbens, 2020; Imai and Kim, 2019). We go beyond this conventional interpretative role by using the DAG to organize the construction of the proposed shrinkage estimator. This construction proceeds in several steps. First, we construct a DAG that provides a graphical representation of the data-generating process and the dependence relationships among the observed outcomes, heterogeneous effects, latent group memberships, and network structure. The corresponding regression model can then be represented through a structural model consistent with this graphical structure. Second, the DAG implies a factorization of the joint distribution into a sequence of lower-dimensional conditional distributions according to the local dependence relationships encoded by the graph. This probabilistic representation provides a Bayesian-network formulation of the model and makes explicit the components of the joint distribution that are relevant for inference on the heterogeneous effects. Third, we treat the posterior mean of the heterogeneous parameters as a class of shrinkage estimators. By Bayes' rule, their posterior distribution is proportional to the product of the likelihood implied by the Bayesian network. Finally, machine-learning methods are employed to learn the latent network group structure and thereby reduce the otherwise high-dimensional prior specification to a collection of group-specific distributions. Conditional on the estimated group memberships, the resulting posterior combines the information contained in the regression likelihood with that contained in the group-specific priors, yielding a tractable empirical-Bayes shrinkage estimator.

The fourth contribution of this paper is to develop a specific machine-learning-assisted empirical Bayes (EB) estimator for the proposed model. When mobility or network connectivity is limited, individual effects may be estimated from relatively little independent information, leading to substantial estimation uncertainty (Jochmans and Weidner, 2019). We exploit shrinkage to trade bias for variance reduction and the latent group structure to assist with the estimation. The presence of latent group structure does not invalidate the least-squares estimator; rather, it provides additional information about cross-sectional heterogeneity that unrestricted least squares does not exploit. Building on the DAG-assisted empirical-Bayes framework introduced above, the resulting estimator combines the regression likelihood with these learned group-specific priors, shrinking each individual effect toward the distribution associated with its estimated group rather than toward a common population-wide distribution. Such selective pooling is particularly useful when limited network connectivity and systematic group-level matching coexist, as

it exploits similarities among observational units while preserving heterogeneity both across and within latent groups.

A further contribution is to characterize how network structure affects the relative performance of least-squares and empirical Bayes estimators through a spectral decomposition of their estimation error. For the unsigned incidence matrix B , let the matrix $L = B^\top B$ denote the graph Laplacian of the underlying bipartite graph. Its eigenvalues describe the information available for estimating contrasts between individual effects: small eigenvalues correspond to weakly connected directions in which least-squares estimates can have large sampling variances. Weak connections can generate such directions, while the identity $\text{tr}(L) = 2n$, where n is the number of observed links, relates the total eigenvalue mass to network size and density. In addition, when latent group structure generates internally well-connected communities with relatively weak connections between them, it can produce small positive Laplacian eigenvalues. These eigenvalues identify directions in which least-squares estimates are imprecise and shrinkage can provide substantial variance reduction. In particular, as an eigenvalue approaches zero, the variance by least-square estimator in the corresponding direction diverges. However, with a fixed positive shrinkage parameter and bounded deviation from the prior mean in that direction, the shrinkage estimator's squared bias remains bounded and its variance is controlled. This analysis explains why informative group-specific priors can improve estimation in weakly connected networks with group structure, while also identifying circumstances in which excessive shrinkage can outweigh the variance reduction.

In theoretical part, our analysis examines the conditions needed for this procedure to provide reliable estimates. We discuss identification of the two-way effects and group-specific prior parameters, and study conditions under which estimated memberships and hyperparameters yield an estimator that approximates an oracle with known groups and prior parameters. We also characterize the bias–variance trade-off relative to least squares. The scope for improvement depends jointly on how informative the group distributions are and how precisely the network identifies individual effects; shrinkage need not improve estimation for every realization of heterogeneity. A Monte Carlo study compares the proposed estimator with unrestricted least squares under a group-structured network design.

We illustrate the method using ratings in the Industrial and Scientific category of Amazon Reviews'23, with 412,947 ratings from 50,985 users on 25,848 products. The application asks whether observed differences in product ratings reflect persistent product-specific components,

differences in reviewers’ rating tendencies, or both. This distinction matters because products reviewed by relatively critical users may receive lower average ratings even when their underlying product-specific components are comparable. Using only the pattern of user–product interactions, the clustering procedure selects 10 user groups and 11 product groups. The estimated prior means differ by approximately 0.315 rating points across the highest- and lowest-mean user groups and by 0.300 points across the corresponding product groups, on a one-to-five scale. These results suggest that who reviews a product contains information relevant to interpreting its ratings. Differences in within-group dispersion also show that network groups are not homogeneous: group membership provides a useful reference distribution while leaving room for substantial individual variation.

For producers, separating reviewer tendencies from product-specific rating components provides a more informative benchmark for interpreting customer feedback. A product with a low average rating but a comparatively favorable adjusted component may attract unusually critical reviewers, whereas a low adjusted component indicates that its evaluations remain relatively unfavorable after accounting for persistent reviewer tendencies. This distinction could help producers identify products warranting closer examination through review text, returns, or quality assessments. For Amazon, the same decomposition could support comparisons among products reviewed by different user populations, while group-informed shrinkage could stabilize estimates for products with limited rating information. The differences in group dispersions further imply that the appropriate degree of pooling can vary across the network. These findings motivate evaluating adjusted rating summaries and their predictive performance; they do not establish improvements in sales, recommendations, or consumer welfare. The estimated components describe conditional patterns in observed ratings rather than causal measures of product quality.

This paper builds on several strands of the literature. First, it relates to the literature on econometric models with latent group structure, in which parameters of interest are represented through a finite number of groups; see, among others, [Su et al. \(2016\)](#), [Bonhomme et al. \(2022\)](#), [Guðmundsson and Brownlees \(2021\)](#), and [Zhang et al. \(2019\)](#). Second, the paper builds on the empirical-Bayes and shrinkage literature, which uses information pooling to improve the estimation of high-dimensional heterogeneous parameters; relevant contributions include [Robbins \(1992\)](#), [Efron \(2012\)](#), [Ignatiadis and Wager \(2019\)](#), and [Efron \(2019\)](#). Third, our methodological framework draws on the literature on directed acyclic graphs, structural causal models, and Bayesian networks, which provides graphical and probabilistic representations of dependence

structures and joint distributions; see, for example, [Pearl et al. \(2003\)](#), [Pearl \(2022\)](#), and [Koller \(2009\)](#). Finally, the paper is related to the growing literature on econometric and statistical models for network data, including [Zhu et al. \(2017\)](#), [Auerbach \(2022\)](#), and [Chen et al. \(2021\)](#). Our contribution lies at the intersection of these literatures by combining two-way heterogeneity, latent group structure in bipartite networks, machine-learning-based group recovery, and group-specific empirical-Bayes shrinkage within a unified estimation framework.

The remainder of the paper is organized as follows. Section 2 presents the model. Section 3 develops the machine-learning-assisted empirical-Bayes estimation procedure. Section 4 establishes the theoretical properties of the estimator. Sections 5 and 6 present, respectively, the simulation study and the empirical application. Section 7 concludes. Additional results and proofs are collected in Appendices A and B.

2 Network Two-Way Effect Regression Model with Group Structure

2.1 Main Model

In this subsection, we introduce the model. Let $\mathcal{G}_n = (\mathcal{S}_n, \mathcal{T}_n, \mathcal{E}_n)$ denote an observed bipartite network, where $\mathcal{S}_n$ and $\mathcal{T}_n$ are the two disjoint sets of nodes and $\mathcal{E}_n$ denotes the set of observed links. Let

$$N = |\mathcal{S}_n|, \quad M = |\mathcal{T}_n|, \quad n = |\mathcal{E}_n|.$$

For every observed link $(i, j) \in \mathcal{E}_n$, consider the two-way effect model for matched data

$$y_{ij} = \alpha_i^s + \alpha_j^t + x'_{ij}\beta + \varepsilon_{ij},$$

where α_i^s denotes the heterogeneous effect associated with node $i \in \mathcal{S}_n$, α_j^t denotes the heterogeneous effect associated with node $j \in \mathcal{T}_n$, x_{ij} is a vector of observed covariates, and ε_{ij} is an idiosyncratic disturbance.

Stacking observations over all links gives

$$Y = B\alpha + X\beta + \varepsilon, \tag{1}$$

where

$$\alpha = \begin{pmatrix} \alpha^s \\ \alpha^t \end{pmatrix} = \left(\alpha_1^s, \dots, \alpha_N^s, \alpha_1^t, \dots, \alpha_M^t \right)',$$

and B is the $n \times (N + M)$ bipartite incidence matrix. Each row of B corresponds to one observed link and contains a one in the column associated with its $\mathcal{S}_n$ -side node and a one in the column associated with its $\mathcal{T}_n$ -side node. Thus, if observation e corresponds to the link (i, j) ,

$$(B\alpha)_e = \alpha_i^s + \alpha_j^t.$$

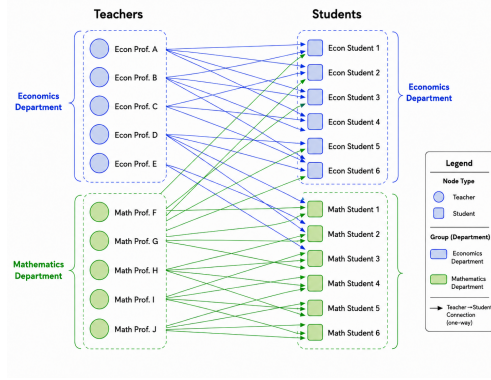
Following the network two-way effect representation, define

$$\tilde{Y} = Y - X\beta.$$

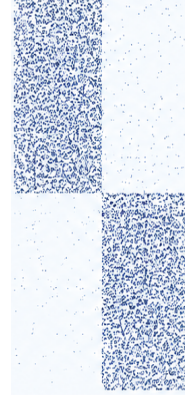
Conditional on β , the model becomes

$$\tilde{Y} = B\alpha + \varepsilon. \tag{2}$$

We assume that the incidence matrix B exhibits a latent group structure. Since B is the matrix representation of the underlying bipartite network, we use B and the network interchangeably throughout the paper. Figure 1 illustrates this structure. Panel (a) shows a bipartite network in which units interact more frequently with units from corresponding groups—for example, economics students primarily take courses taught by economics instructors, while maintaining fewer links with instructors from other departments. Panel (b) presents the corresponding adjacency matrix after reordering nodes by group membership, revealing a clear block structure associated with the latent groups on both sides of the network.



(a)



(b)

Figure 1: Panel (a): a bipartite network with latent group structure on both sides. Panel (b): the heatmap of the corresponding adjacency matrix after reordering nodes by group membership, showing the resulting block structure.

Furthermore, we assume that the two sets of heterogeneous effects possess potentially different latent group structures. In particular, suppose that the $\mathcal{S}_n$ side contains K latent groups and the $\mathcal{T}_n$ side contains L latent groups. Let

$$g_i \in \{1, \dots, K\}, \quad h_j \in \{1, \dots, L\}$$

denote the respective memberships.

Rather than imposing a common fixed effect on all units belonging to the same group, we assume that group membership determines the distribution of individual heterogeneity. Specifically,

$$\begin{aligned} \alpha_i^s \mid g_i = k &\sim \mathcal{N}(\mu_k^s, \tau_{k,s}^2), \quad k = 1, \dots, K, \\ \alpha_j^t \mid h_j = \ell &\sim \mathcal{N}(\mu_\ell^t, \tau_{\ell,t}^2), \quad \ell = 1, \dots, L. \end{aligned}$$

Hence units belonging to the same group share the same prior distribution but are not required to have identical heterogeneous effects. Equivalently, we may write

$$\begin{aligned} \alpha_i^s &= \mu_{g_i}^s + u_i^s, & \mathbb{E}[u_i^s \mid g_i] &= 0, \\ \alpha_j^t &= \mu_{h_j}^t + u_j^t, & \mathbb{E}[u_j^t \mid h_j] &= 0. \end{aligned}$$

Collect the corresponding prior means and variances for both sides into the vectors

$$\mu = \left(\mu_{g_1}, \dots, \mu_{g_N}, \mu_{h_1}, \dots, \mu_{h_M} \right)',$$

and

$$\tau^2 = \left(\tau_{g_1}^2, \dots, \tau_{g_N}^2, \tau_{h_1}^2, \dots, \tau_{h_M}^2 \right)'.$$

In addition to the main specification, we propose a directed acyclic graph (DAG) that represents the underlying data-generating mechanism. Figure 2 illustrates the proposed generative structure and the associated dependence relationships among the model components. The DAG provides a graphical representation of the relationships among the observed outcomes $\tilde{Y}$, network links B , latent effects α , group memberships Z , and prior parameters (μ, τ^2) .

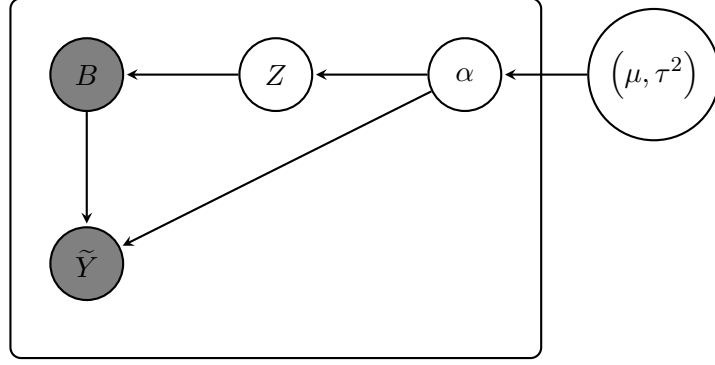


Figure 2: DAG assists with proposed model. Circle outside the plate denote hyperparameters, whereas circles inside the plates represent latent or observed variables. Shaded (gray) circles indicate observed quantities. In particular, the network B and the outcome $\tilde{Y}$ are observed.

A directed acyclic graph can be associated with both a structural-equation representation and a probabilistic graphical representation. When the two representations share the same DAG, the structural equations describe how each variable is generated from its parent variables and an exogenous innovation, whereas the corresponding Bayesian network expresses the joint distribution through conditional distributions implied by the same parent-child relationships. In the present paper, we use this connection as an organizational device rather than relying on the intervention-based causal interpretation of a structural causal model. In particular, Figure 2 can first be represented by a system of structural equations describing the dependence relationships among the latent group memberships, heterogeneous effects, network structure, and observed outcome. The same DAG then implies a factorization of the joint distribution of these variables into a product of lower-dimensional conditional distributions. This factorization is useful for esti-

mation because the posterior distribution of the heterogeneous effects can be obtained from the components of the joint distribution that depend on those effects. Consequently, the DAG plays two roles in our framework: it provides a graphical representation of the dependence structure of the model and, at the same time, organizes the probabilistic factorization that leads to the proposed empirical-Bayes estimator.

Formally, we introduce the following structural representation.

Definition 1. *Consider the model in equation (2) together with the probabilistic graph in Figure 2. The variables admit the structural representation*

$$\begin{aligned}\alpha &= f_\alpha(\varepsilon_\alpha), \\ Z &= f_Z(\alpha, \varepsilon_Z), \\ B &= f_B(Z, \varepsilon_B), \\ \tilde{Y} &= f_{\tilde{Y}}(B, \alpha, \varepsilon),\end{aligned}$$

where $\varepsilon_\alpha, \varepsilon_Z, \varepsilon_B, \varepsilon$ denote exogenous innovations. The outcome equation is specified as

$$f_{\tilde{Y}}(B, \alpha, \varepsilon) = B\alpha + \varepsilon_{\tilde{Y}},$$

and

$$f_\alpha = \varepsilon_\alpha,$$

where $\varepsilon_\alpha \sim \mathcal{N}(\mu, \text{diag}(\tau^2))$, while $f_Z(\cdot)$ and $f_B(\cdot)$ are left unspecified.

The structural representation in Definition 1 induces the Bayesian-network factorization

$$\mathbb{P}(\tilde{Y}, B, \alpha, Z) = \mathbb{P}(\alpha) \mathbb{P}(Z | \alpha) \mathbb{P}(B | Z) \mathbb{P}(\tilde{Y} | B, \alpha). \quad (3)$$

Conditional on B and $\tilde{Y}$, we have posterior,

$$\mathbb{P}(\alpha | \tilde{Y}, B) \propto \sum_Z \mathbb{P}(\tilde{Y}, B, \alpha, Z). \quad (4)$$

We first insert equation (3) into equation (4). Next, in the feasible implementation, the latent membership Z is replaced by its machine-learning estimate $\hat{Z}$. Then, a sequence of conditional distributions can be further simplified into two terms, where the first term is the likelihood im-

plied by the regression equation, whereas the second term is the group-specific prior distribution of the heterogeneous effects. Furthermore, the unknown group-specific hyperparameters are estimated from the data. This posterior mean

$$\hat{\alpha}^{EB} = \mathbb{E} \left[\alpha \mid \tilde{Y}, B, \hat{Z} \right]$$

defines the proposed machine-learning-assisted empirical-Bayes shrinkage estimator. The details of the estimation procedure are deferred to the next section. In the remainder of this section, we characterize the conditional distributions implied by the structural equations in Definition 1.

We explicitly specify the structural functions f_α and $f_{\tilde{Y}}$, since these two components are sufficient for constructing the empirical-Bayes estimator, while leaving f_Z and f_B unspecified to preserve flexibility in the latent group formation and network-generation mechanisms. The structural equations in Definition 1 induce corresponding conditional distributions. For example, under

$$\tilde{Y} = B\alpha + \varepsilon, \quad \varepsilon \sim \mathcal{N} \left(0, \sigma_\varepsilon^2 I_n \right),$$

implies

$$\tilde{Y} \mid B, \alpha \sim \mathcal{N} \left(B\alpha, \sigma_\varepsilon^2 I_n \right).$$

Likewise, under the Gaussian group-specific prior,

$$\alpha \sim \mathcal{N} \left(\mu, D(\tau^2) \right),$$

where

$$D(\tau^2) = \text{diag} \left(\tau_{g_1}^2, \dots, \tau_{g_N}^2, \tau_{h_1}^2, \dots, \tau_{h_M}^2 \right),$$

denotes the diagonal matrix of prior variances, and μ denotes the vector of prior means as defined above. Integrating out α yields

$$\tilde{Y} \mid B, Z \sim \mathcal{N} \left(B\mu, BD(\tau^2)B' + \sigma_\varepsilon^2 I_n \right).$$

Hence, the structural functions f_α and $f_{\tilde{Y}}$ directly determine the prior and likelihood components required for empirical-Bayes estimation. By contrast, because f_Z and f_B are left unspecified, the corresponding distributions governing latent group formation and network generation are also left unrestricted. In particular, under the DAG direction $\alpha \rightarrow Z$ and $Z \rightarrow B$, the distributions

$\mathbb{P}(Z \mid \alpha)$ and $\mathbb{P}(B \mid Z)$ remain unspecified, whereas the prior $\mathbb{P}(\alpha)$ and the regression likelihood $\mathbb{P}(\tilde{Y} \mid B, \alpha)$ are specified explicitly.

3 Machine-Learning-Assisted Empirical Bayes Estimation

The core component of the estimation procedure is the empirical Bayes estimator, defined as the posterior mean of the unit-level heterogeneity. To derive this estimator, we begin with the joint distribution implied by the graphical model in Figure 2. Omitting the hyperparameters from the notation for brevity, the joint distribution in equation (3) can be decomposed further as follows:

$$\begin{aligned} \mathbb{P}(\alpha \mid \tilde{Y}, B) &\propto \sum_Z \mathbb{P}(\tilde{Y}, B, Z, \alpha) \\ &= \sum_Z \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(B \mid Z) \mathbb{P}(Z \mid \alpha) \mathbb{P}(\alpha) \end{aligned} \quad (5)$$

$$\propto \sum_Z \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(Z \mid B) \mathbb{P}(\alpha \mid Z), \quad (6)$$

$$\begin{aligned} &\text{conditional on estimating } Z \text{ such that } \mathbb{P}(\hat{Z} \mid B) \rightarrow 1, \\ &\rightarrow \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \hat{Z}). \end{aligned} \quad (7)$$

Suppose further that the latent group structure can be consistently recovered from the observed network B , in the sense that

$$\mathbb{P}(\hat{Z} = Z \mid B) \rightarrow 1.$$

Under this condition, posterior uncertainty over the group memberships vanishes asymptotically, and equation (6) reduces to the feasible representation in equation (7), with the unknown Z replaced by $\hat{Z}$. The proposed empirical Bayes estimator is then constructed as the posterior mean

$$\hat{\alpha}^{EB} = \mathbb{E}(\alpha \mid \tilde{Y}, B, \hat{Z}).$$

Define the vector of group-specific prior means as

$$\mu_g = \left(\mu_1^s, \dots, \mu_K^s, \mu_1^t, \dots, \mu_L^t \right)',$$

and the vector of group-specific prior variances as

$$\tau_g^2 = \left((\tau_1^s)^2, \dots, (\tau_K^s)^2, (\tau_1^t)^2, \dots, (\tau_L^t)^2 \right)'.$$

Define the membership matrices

$$Z_s \in \{0, 1\}^{N \times K}, \quad Z_t \in \{0, 1\}^{M \times L},$$

where every row contains exactly one nonzero element indicating the group membership for unit i : for instance, if unit i belongs to group 2 out of three groups, the corresponding row of Z_s is $(0, 1, 0)$. The joint membership matrix is

$$Z = \begin{pmatrix} Z_s & 0 \\ 0 & Z_t \end{pmatrix}.$$

The vector of unit-specific prior means, $\mu = (\mu_{g_1}^s, \dots, \mu_{g_N}^s, \mu_{h_1}^t, \dots, \mu_{h_M}^t)'$, can be written compactly as $\mu = Z\mu_g$. Note that $N + M \gg K + L$, where K is the number of groups among the N units on the first side and L is the number of groups among the M units on the second side. An analogous representation applies to the unit-specific prior variances: $\tau^2 = Z\tau_g^2$, where $\tau^2 = (\tau_{g_1}^2, \dots, \tau_{g_N}^2, \tau_{h_1}^2, \dots, \tau_{h_M}^2)'$. Hence, conditional on the group memberships Z , the empirical Bayes estimation problem reduces from estimating the $(N + M)$ -dimensional vectors of unit-specific prior parameters μ and τ^2 to estimating the $(K + L)$ -dimensional vectors of group-specific hyperparameters μ_g and τ_g^2 .

3.1 Detecting Z Using Machine Learning

In this subsection, we discuss methods for estimating the latent membership matrix Z from the observed network B , and we denote the resulting estimator by $\hat{Z}$. Estimating Z is an important step in constructing Equation (7). Under the conditional-independence assumptions of the model, the plug-in posterior based on the estimated memberships satisfies

$$\mathbb{P}(\alpha \mid \tilde{Y}, B, \hat{Z}) \propto \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \hat{Z}),$$

where $\mathbb{P}(\tilde{Y} \mid \alpha, B)$ is the likelihood and $\mathbb{P}(\alpha \mid \hat{Z})$ is the group-specific prior given estimated group membership.

Given B , our objective is therefore to apply a network-clustering method that consistently estimates the latent group assignments. Because group labels are identifiable only up to permutation, an appropriate exact-recovery condition is

$$\mathbb{P}(\hat{Z} = ZP \text{ for some } P \in \mathcal{P}) \rightarrow 1,$$

where $\mathcal{P}$ denotes the set of admissible group-label permutation matrices.

Our framework can accommodate any group-detection algorithm satisfying the recovery condition and convergence rate required for the subsequent empirical Bayes analysis. One such approach is spectral clustering, which is among the most widely used and computationally efficient methods for detecting latent group structure in networks (Ng et al., 2001; Rohe et al., 2011, 2016). In the bipartite setting, the left and right singular vectors of a suitably normalized bi-adjacency matrix provide spectral embeddings for the two sets of nodes, which can then be clustered separately to estimate their respective memberships.

3.2 The Estimator

In this subsection, we construct the empirical Bayes estimator based on equation (7). We first describe the procedure and then dive into the detail.

The first step is to estimate the numbers of latent groups on the two sides of the network, denoted by K and L , respectively. These quantities determine the numbers of group-specific prior distributions: K distributions for the effects on one side of the network and L distributions for those on the other. Various procedures have been proposed for selecting the number of network communities (Le and Levina, 2015; Li et al., 2022; Karrer and Newman, 2011; Gao et al., 2018). Because community-number selection is not the primary focus of this paper, our framework permits any computationally feasible and theoretically justified selection procedure.

The second step is to estimate the slope coefficients β from a preliminary two-way effect regression that controls for the node effects represented by B . In particular, letting

$$M_B = I - B(B'B)^\dagger B',$$

where $(B'B)^\dagger$ denotes the Moore–Penrose generalized inverse, we compute

$$\hat{\beta} = (X'M_B X)^{-1} X'M_B Y$$

and construct the residualized outcome

$$\tilde{Y} = Y - X\hat{\beta}.$$

Here, β is a nuisance parameter estimated in the preliminary stage, while X is an observed covariate matrix.

The third step is to estimate the latent membership matrix Z from the network B , yielding $\hat{Z}$. Any network-clustering procedure is admissible provided that it satisfies the recovery condition required by the subsequent empirical Bayes analysis. Although our implementation uses spectral clustering, the empirical Bayes framework itself does not depend on a particular clustering algorithm.

Finally, given $\tilde{Y}$, B , and $\hat{Z}$, we construct the empirical Bayes estimator. Under the Gaussian prior specification, we first estimate the group-level hyperparameters

$$\lambda = (\mu_g, \tau_g^2)$$

by maximizing the marginal likelihood in Equation (11). Denoting the resulting marginal maximum-likelihood estimator by $\hat{\lambda}$, we substitute $\hat{\lambda}$ and $\hat{Z}$ into the posterior-mean expression in Equation (10). The resulting feasible posterior mean is the final empirical Bayes estimate of the heterogeneous node effects produced by Algorithm 1.

Next, we discuss the estimation details. The covariance matrix of α conditional on Z is

$$D(Z\tau_g^2) = \text{diag}(Z\tau_g^2).$$

Consequently,

$$\alpha \mid Z \sim \mathcal{N}\left(Z\mu_g, D(Z\tau_g^2)\right).$$

For the Gaussian benchmark, assume

$$\varepsilon \mid (B, X) \sim \mathcal{N}\left(0, \sigma_\varepsilon^2 I_n\right).$$

It follows that the marginal distribution of the residualized outcome is

$$\tilde{Y} \mid B, Z \sim \mathcal{N}\left(BZ\mu_g, BD\left(Z\tau_g^2\right)B' + \sigma_\varepsilon^2 I_n\right). \quad (8)$$

Equation (8) provides the basis for both identification of the group-specific hyperparameters and their marginal maximum-likelihood estimation.

Consequently, conditional on the estimated group structure $\hat{Z}$, the likelihood and prior jointly imply the following multivariate normal structure.

$$\begin{bmatrix} \tilde{Y} \\ \alpha \end{bmatrix} \mid B, \hat{Z} \sim \mathcal{N}\left(\begin{bmatrix} B\hat{Z}\mu_g \\ \hat{Z}\mu_g \end{bmatrix}, \begin{bmatrix} BD\left(\hat{Z}\tau_g^2\right)B' + \sigma_\varepsilon^2 I_n & BD\left(\hat{Z}\tau_g^2\right) \\ D\left(\hat{Z}\tau_g^2\right)B' & D\left(\hat{Z}\tau_g^2\right) \end{bmatrix}\right). \quad (9)$$

Using standard results for conditional distributions of jointly normal random variables, the empirical Bayes estimator $\hat{\alpha} = \mathbb{E}(\alpha \mid B, \tilde{Y}, \hat{Z})$ admits the following closed-form expression:

$$\hat{Z}\mu_g + BD\left(\hat{Z}\tau_g^2\right)\left(BD\left(\hat{Z}\tau_g^2\right)B' + \sigma_\varepsilon^2 I_n\right)^{-1}\left(\tilde{Y} - B\hat{Z}\mu_g\right). \quad (10)$$

Then, given $\hat{Z}$, the group-level parameters μ_g and τ_g can be estimated via marginal maximum likelihood using equation (11).

$$\mathbb{P}(\tilde{Y} \mid B, \hat{Z}, \lambda) = \int_{\alpha} \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \hat{Z}, \lambda) d\alpha, \quad (11)$$

which give us the estimated parameters $\hat{\lambda} = (\mu_g, \tau_g)$ as follows,

$$\hat{\lambda}^{MLE} = \arg \max_{\lambda \in \mathcal{J}} \mathbb{P}(\tilde{Y} \mid B, \hat{Z}, \lambda). \quad (12)$$

Substituting the marginal maximum likelihood estimates $\hat{\lambda}^{MLE}$ into equation (10) yields the empirical Bayes estimator. We summarize the estimation procedure in the following algorithm.

Algorithm 1 ML-Assisted Empirical Bayes Estimation

Input: outcome vector Y , covariate matrix X , and incidence matrix B .

- 1: Determine the number of groups $\widehat{K}$ and $\widehat{L}$ in B using a suitable group-selection procedure.
- 2: Using OLS, estimate $\widehat{\beta}$ and calculate $\widetilde{Y} = Y - X\widehat{\beta}'$.
- 3: Detect the group membership structure $\widehat{Z}$ in B .
- 4: Construct the EB estimator $\mathbb{E}(\alpha \mid B, \widetilde{Y}, \widehat{Z}, \lambda)$, where $\lambda = (\mu_g, \tau_g^2)$.
- 5: Estimate $\widehat{\lambda}$ in $\mathbb{P}(\widetilde{Y} \mid B, \widehat{Z}, \lambda) = \int_{\alpha} \mathbb{P}(\widetilde{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \widehat{Z}, \lambda) d\alpha$.
- 6: Substitute $\widehat{\lambda}$ into the EB estimator.

Output: ML-Assisted Empirical Bayes estimator $\widehat{\alpha}^{EB} = \mathbb{E}(\alpha \mid B, \widetilde{Y}, \widehat{Z}, \widehat{\lambda})$.

4 Theory

This section develops the identification conditions and theoretical properties of the proposed machine-learning-assisted empirical Bayes (ML-EB) estimator. The analysis proceeds in four steps. First, we derive conditions for identifying the two-way heterogeneous effects and the associated group-specific prior parameters in the bipartite-network setting. Second, we provide conditions under which the latent group memberships on both sides of the network can be consistently recovered. Third, we establish consistency of the estimated prior hyperparameters and show that, under consistent group recovery and hyperparameter estimation, the feasible empirical Bayes estimator asymptotically approaches its oracle counterpart. Finally, we characterize the mean squared error of the proposed estimator and use this characterization to clarify its bias–variance trade-off relative to the conventional least-squares estimator.

4.1 Identification

Identification of the proposed two-way effect model relies on two main conditions: an appropriate location normalization and connectedness of the observed bipartite network. Together, these conditions ensure that the two sets of unobserved heterogeneous effects are uniquely identified relative to the chosen normalization.

A fundamental identification issue in the two-way effect model is location indeterminacy. To

see this, consider the residualized equation

$$\tilde{y}_{ij} = \alpha_i^s + \alpha_j^t + \varepsilon_{ij}.$$

For any scalar $c \in \mathbb{R}$, define

$$\alpha_i^{s,*} = \alpha_i^s + c, \quad \alpha_j^{t,*} = \alpha_j^t - c. \quad (13)$$

Then

$$\alpha_i^{s,*} + \alpha_j^{t,*} = \alpha_i^s + \alpha_j^t,$$

so the transformation in equation (13) leaves every conditional mean unchanged.

Define

$$v_0 = \begin{pmatrix} \mathbf{1}_N \\ -\mathbf{1}_M \end{pmatrix}. \quad (14)$$

Since each row of B contains one incidence from each side of the bipartite network, $Bv_0 = 0$.

Hence v_0 belongs to the null space of B . A location normalization is therefore necessary.

Assumption 1 (Location normalization). *The group means on the $\mathcal{T}_n$ side satisfy*

$$\sum_{\ell=1}^L \pi_\ell^t \mu_\ell^t = 0,$$

where

$$\pi_\ell^t = \frac{M_\ell}{M}, \quad M_\ell = \sum_{j=1}^M \mathbf{1}\{h_j = \ell\}.$$

Moreover,

$$\mathbb{E} \left[\alpha_j^t - \mu_{h_j}^t \mid h_j \right] = 0.$$

The normalization could equivalently be imposed on the $\mathcal{S}_n$ side without affecting fitted values. Assumption 1 fixes the relative location of the two sets of heterogeneous effects.

The second identification condition concerns the topology of the observed network. Let C_n denote the number of connected components of $\mathcal{G}_n$. For the bipartite incidence matrix used above,

$$\text{rank}(B) = N + M - C_n.$$

Thus, every connected component generates one location indeterminacy.

Assumption 2 (Network connectivity). *The bipartite graph $\mathcal{G}_n$ is connected, so that*

$$C_n = 1$$

and consequently

$$\text{rank}(B) = N + M - 1.$$

Every node included in the analysis has strictly positive degree.

In connectedness, the null space of B is one-dimensional and is spanned by v_0 . Hence the location normalization removes the only remaining indeterminacy.

Proposition 1 (Identification of the normalized two-way effects). *Suppose Assumptions 1–2 hold. Conditional on β , let α_1 and α_2 satisfy the same location normalization. If*

$$B\alpha_1 = B\alpha_2,$$

then

$$\alpha_1 = \alpha_2.$$

Thus the normalized two-way heterogeneous effects are point identified.

Remark 1 (Group Mean). *Importantly, introducing group-specific priors does not by itself remove this location indeterminacy when the prior means are estimated from the same data. In particular, consider the transformation*

$$\begin{aligned} \alpha_i^{s,*} &= \alpha_i^s + c, & \text{and} & & \mu_k^{s,*} &= \mu_k^s + c, \\ \alpha_j^{t,*} &= \alpha_j^t - c, & \text{and} & & \mu_\ell^{t,*} &= \mu_\ell^t - c. \end{aligned}$$

The likelihood is unchanged, while

$$\alpha_i^{s,*} - \mu_{g_i}^{s,*} = \alpha_i^s - \mu_{g_i}^s \quad \text{and} \quad \alpha_j^{t,*} - \mu_{h_j}^{t,*} = \alpha_j^t - \mu_{h_j}^t.$$

Thus the group-prior density is also invariant. The empirical Bayes prior therefore provides regularization and partial pooling, but the location normalization remains necessary for identification.

The covariates must contain variation that is not perfectly explained by the two sets of node effects. Let P_B denote the projection matrix onto the identified column space of B , and define

$$M_B = I_n - P_B.$$

Assumption 3 (Covariate identification). *Let $p = \dim(\beta)$. The residualized design matrix has full column rank:*

$$\text{rank}(M_B X) = p.$$

Moreover, for some constant $c_X > 0$,

$$\lambda_{\min} \left(\frac{1}{n} X' M_B X \right) \geq c_X$$

with probability approaching one.

Assumption 3 excludes covariates that are perfectly collinear with the two-way heterogeneous effects. Under this condition, β is separately identified from α .

Remark 2 (Common intercept). *The baseline specification in (1) does not require a separate unrestricted common intercept because a constant vector lies in the span generated by the two sets of unrestricted effects. If the model is instead parameterized as*

$$y_{ij} = \beta_0 + \alpha_i^s + \alpha_j^t + x'_{ij}\beta + \varepsilon_{ij},$$

then two location restrictions are required, for example

$$\sum_i w_i^s \alpha_i^s = 0, \quad \sum_j w_j^t \alpha_j^t = 0,$$

so that the common intercept β_0 is uniquely defined.

Conditional on the latent memberships $\mathbb{E}[\tilde{y}_{ij} \mid g_i = k, h_j = \ell] = \mu_k^s + \mu_\ell^t$, identification of the group means itself has a two-way structure.

Define a group-level bipartite graph whose first partition consists of the K groups on the $\mathcal{S}_n$ side and whose second partition consists of the L groups on the $\mathcal{T}_n$ side. Groups k and ℓ are connected if

$$\mathbb{P}((i, j) \in \mathcal{E}_n \mid g_i = k, h_j = \ell) > 0.$$

Assumption 4 (Group-level connectedness). *The induced (K, L) group-level bipartite graph is connected and every group contains at least one observed node.*

Under Assumptions 1 and 4, the group means

$$\mu_1^s, \dots, \mu_K^s, \mu_1^t, \dots, \mu_L^t$$

are identified from the first conditional moment. When the individual-level graph is connected and all groups are nonempty, its contraction into groups is also connected, so Assumption 4 is typically implied by Assumption 2.

Identification of the group-specific prior variances requires information from second moments. Let

$$\vartheta = \left(\tau_{1,s}^2, \dots, \tau_{K,s}^2, \tau_{1,t}^2, \dots, \tau_{L,t}^2, \sigma_\varepsilon^2 \right)'.$$

For $r = 1, \dots, K + L$, let D_r be a diagonal matrix that selects the nodes belonging to prior group r . Then

$$D(\tau^2) = \sum_{r=1}^{K+L} \tau_r^2 D_r,$$

and the marginal covariance matrix can be written as

$$\Omega(\vartheta) = \sum_{r=1}^{K+L} \tau_r^2 B D_r B' + \sigma_\varepsilon^2 I_n.$$

Assumption 5 (Variance-component identification). *The matrices*

$$\{B D_1 B', \dots, B D_{K+L} B', I_n\}$$

are linearly independent in the population limit. That is,

$$a_0 I_n + \sum_{r=1}^{K+L} a_r B D_r B' = 0$$

implies

$$a_0 = a_1 = \dots = a_{K+L} = 0.$$

Assumption 5 implies that distinct combinations of group-specific heterogeneity variances and idiosyncratic noise cannot generate the same covariance matrix. The economic intuition is

particularly transparent from repeated links. For example, under conditional independence,

$$\text{Cov}(\tilde{y}_{ij}, \tilde{y}_{ij'} \mid g_i = k) = \tau_{k,s}^2, \quad j \neq j',$$

while

$$\text{Cov}(\tilde{y}_{ij}, \tilde{y}_{i'j} \mid h_j = \ell) = \tau_{\ell,t}^2, \quad i \neq i',$$

Hence repeated observations sharing a node on either side of the network provide information that separates the corresponding heterogeneity variance from the idiosyncratic error variance. In contrast, a network in which all nodes have degree one generally provides insufficient covariance information for separating all variance components.

4.2 Recovery of the latent group structure

Given identification, consistency of the feasible empirical Bayes estimator additionally requires consistent recovery of the latent group memberships and sufficient within-group information for consistent estimation of the group-specific prior means and variances, as well as the error variance.

The latent memberships are unknown and must be recovered from the observed network. Let $\hat{Z}_s$ and $\hat{Z}_t$ denote the estimated memberships obtained from a consistent network clustering procedure, such as spectral clustering or singular-vector-based co-clustering. Because the numerical labels of latent groups are arbitrary, consistency is defined up to label permutations. Let Π_s and Π_t denote permutation matrices of dimensions K and L , respectively.

Assumption 6 (Network community structure). *Conditional on the latent memberships, the bipartite adjacency matrix satisfies a degree-corrected stochastic block model*

$$A_{ij} \mid Z, \theta \sim \text{Bernoulli}(\theta_i^s \theta_j^t P_{g_i h_j}).$$

The numbers of groups K and L are fixed. Moreover, the limiting group proportions are bounded away from zero,

$$\frac{N_k}{N} \rightarrow \pi_k^s > 0, \quad \frac{M_\ell}{M} \rightarrow \pi_\ell^t > 0,$$

and distinct groups have distinct population connection profiles:

$$\min_{k \neq k'} \|\mathbb{P}_{k,\cdot} - \mathbb{P}_{k',\cdot}\| \geq \delta_s > 0,$$

$$\min_{\ell \neq \ell'} \|\mathbb{P}_{\cdot,\ell} - \mathbb{P}_{\cdot,\ell'}\| \geq \delta_t > 0.$$

Together with an appropriate minimum-degree condition, these restrictions allow standard community-detection procedures to consistently recover the latent partitions.

Proposition 2 (Consistency of latent-group recovery). *Suppose Assumption 6 holds and the latent memberships are estimated using a consistent bipartite spectral clustering procedure. Then there exist permutation matrices Π_s and Π_t such that*

$$\frac{1}{N} \|\widehat{Z}_s - Z_s \Pi_s\| \xrightarrow{p} 0,$$

$$\frac{1}{M} \|\widehat{Z}_t - Z_t \Pi_t\| \xrightarrow{p} 0.$$

Equivalently,

$$d_Z(\widehat{Z}, Z) \xrightarrow{p} 0,$$

up to a permutation of the group labels.

This result provides the formal justification for treating the recovered network communities as group-specific prior information in the second stage of the proposed ML-EB procedure.

4.3 Estimation and consistency of the prior hyperparameters

Let

$$\psi = (\mu', (\tau^2)', \sigma_\varepsilon^2)'$$

collect the unknown prior and disturbance parameters. Conditional on B and Z , define

$$m_\psi = BZ\mu,$$

$$\Omega_\psi = BD(\tau^2)B' + \sigma_\varepsilon^2 I_n.$$

From (11), the marginal log likelihood, up to a constant, is

$$\begin{aligned}\ell_n(\psi; Z) = & -\frac{1}{2} \log |\Omega_\psi| \\ & -\frac{1}{2} (\tilde{Y} - BZ\mu)' \Omega_\psi^{-1} (\tilde{Y} - BZ\mu).\end{aligned}$$

The feasible empirical Bayes hyperparameter estimator is

$$\hat{\psi} = \arg \max_{\psi \in \Psi} \ell_n(\psi; \hat{Z}).$$

Assumption 7 (Parameter space). *The true parameter ψ_0 is uniquely identified and belongs to the interior of a compact parameter space Ψ . In the regular interior case, there exist constants*

$$0 < \underline{\tau}^2 < \bar{\tau}^2 < \infty$$

and

$$0 < \underline{\sigma}^2 < \bar{\sigma}^2 < \infty$$

such that

$$\begin{aligned}\underline{\tau}^2 &\leq \tau_r^2 \leq \bar{\tau}^2, & r = 1, \dots, K + L, \\ \underline{\sigma}^2 &\leq \sigma_\varepsilon^2 \leq \bar{\sigma}^2.\end{aligned}$$

Allowing $\tau_r^2 = 0$ places the corresponding parameter on the boundary and requires a separate nonstandard asymptotic analysis. We therefore focus on the regular interior case.

Assumption 8 (Group growth). *Recall from Assumption 6 that the limiting group shares satisfy*

$$\frac{N_k}{N} \rightarrow \pi_k^s > 0, \quad \frac{M_\ell}{M} \rightarrow \pi_\ell^t > 0,$$

for every $k = 1, \dots, K$ and $\ell = 1, \dots, L$. Since $N, M \rightarrow \infty$, this condition also implies that every group grows without bound,

$$N_k \rightarrow \infty, \quad M_\ell \rightarrow \infty,$$

which in turn ensures that the group-specific prior means and variances can be consistently estimated as the sample size increases.

Assumption 9 (Uniform convergence of the marginal likelihood). *For some deterministic sequence $a_n \rightarrow \infty$ and a deterministic criterion function $Q(\psi)$,*

$$\sup_{\psi \in \Psi} \left| \frac{1}{a_n} \ell_n(\psi; Z) - Q(\psi) \right| \xrightarrow{p} 0.$$

Moreover, $Q(\psi)$ is continuous on Ψ and is uniquely maximized at ψ_0 .

Proposition 3 (Consistency of the estimated group priors). *Suppose Assumptions 1–9 hold. Suppose further that either exact group recovery holds or the remaining classification error has an asymptotically negligible effect on the normalized marginal likelihood, namely*

$$\sup_{\psi \in \Psi} \left| \frac{1}{a_n} \ell_n(\psi; \hat{Z}) - \frac{1}{a_n} \ell_n(\psi; Z) \right| = o_p(1).$$

Then, up to permutation of group labels,

$$\hat{\psi} \xrightarrow{p} \psi_0.$$

In particular,

$$\begin{aligned} \hat{\mu}_k^s &\xrightarrow{p} \mu_k^s, & \text{and} & & \hat{\tau}_{k,s}^2 &\xrightarrow{p} \tau_{k,s}^2, \\ \hat{\mu}_\ell^t &\xrightarrow{p} \mu_\ell^t, & \text{and} & & \hat{\tau}_{\ell,t}^2 &\xrightarrow{p} \tau_{\ell,t}^2, \end{aligned}$$

and

$$\hat{\sigma}_\varepsilon^2 \xrightarrow{p} \sigma_\varepsilon^2.$$

4.4 Oracle and feasible empirical Bayes estimators

Under the conditions above, the feasible estimator asymptotically approaches the corresponding oracle empirical Bayes estimator that knows the true group assignments and hyperparameters.

Suppose first that the true group structure Z and true hyperparameters ψ_0 are known. Define

$$m = Z\mu, \quad D = D(\tau^2).$$

The joint Gaussian structure implies that the oracle posterior mean of α is

$$\alpha^{EB,O} = m + DB' (BDB' + \sigma_\varepsilon^2 I_n)^{-1} (\tilde{Y} - Bm).$$

The feasible ML-EB estimator replaces Z and ψ_0 by their estimated counterparts:

$$\hat{\alpha}^{EB} = \hat{m} + \hat{D}B' (B\hat{D}B' + \hat{\sigma}_\varepsilon^2 I_n)^{-1} (\tilde{Y} - B\hat{m}), \quad (15)$$

where

$$\hat{m} = \hat{Z}\hat{\mu}, \quad \hat{D} = D(\hat{\tau}^2).$$

Equation (15) makes the shrinkage mechanism explicit. The estimator starts from the group-specific prior mean $\hat{m}$ and updates it using the discrepancy

$$\tilde{Y} - B\hat{m}.$$

The magnitude of the update depends jointly on the within-group heterogeneity variances, the disturbance variance, and the topology of the bipartite network.

Theorem 1 (Oracle equivalence). *Let*

$$p_n = N + M$$

denote the total number of nodes in the bipartite network. Suppose that Assumptions 1-9 hold.

Then, with Proposition 2 and 3, the feasible ML-EB estimator satisfies

$$\frac{1}{p_n} \left\| \hat{\alpha}^{EB} - \alpha^{EB,O} \right\|^2 \xrightarrow{p} 0,$$

where $\alpha^{EB,O}$ denotes the oracle empirical Bayes estimator constructed using the true group memberships and the true hyperparameters.

Theorem 1 implies that estimation of the network groups and prior hyperparameters has an asymptotically negligible effect on the average squared distance between the feasible ML-EB estimator and its infeasible oracle counterpart. Hence the first-stage machine-learning procedure does not alter the limiting second-stage estimator when group recovery and hyperparameter estimation are sufficiently accurate.

4.5 Conditional mean squared error and the bias–variance trade-off

Relative to the conventional two-way least-squares estimator, empirical Bayes introduces shrinkage bias in exchange for a reduction in estimation variance. Accordingly, its conditional mean squared error is lower whenever the variance reduction dominates the increase in squared bias induced by shrinkage. In addition, under a correctly specified group-specific prior and quadratic loss, the posterior mean is the Bayes estimator and therefore minimizes integrated posterior risk. Consequently, its integrated Bayes risk is weakly lower than that of any alternative estimator, including unrestricted least squares, under the maintained prior specification.

The empirical Bayes estimator can also be characterized as a regularized two-way effect estimator. Let

$$m = Z\mu, \quad D = D(\tau^2),$$

and define

$$W = DB' \left(BDB' + \sigma_\varepsilon^2 I_n \right)^{-1}.$$

Then the oracle posterior mean can be written as

$$\hat{\alpha}^{EB} = m + W(\tilde{Y} - Bm).$$

Since

$$\tilde{Y} = B\alpha + \varepsilon,$$

we obtain

$$\begin{aligned} \hat{\alpha}^{EB} - \alpha &= m - \alpha + WB(\alpha - m) + W\varepsilon \\ &= -(I - WB)(\alpha - m) + W\varepsilon. \end{aligned}$$

Conditional on the realized heterogeneous effects α ,

$$\mathbb{E} \left[\hat{\alpha}^{EB} - \alpha \right] = -(I - WB)(\alpha - m).$$

Thus empirical Bayes introduces shrinkage bias whenever $\alpha \neq m$. In contrast,

$$\text{Var} \left(\hat{\alpha}^{EB} \right) = \sigma_\varepsilon^2 WW'.$$

The conditional mean squared error under quadratic loss is therefore

$$\begin{aligned} \text{MSE}_{EB}(\alpha) &= \|(I - WB)(\alpha - m)\|^2 \\ &\quad + \sigma_\varepsilon^2 \text{tr}(WW'). \end{aligned} \tag{16}$$

Equation (16) explicitly decomposes the estimation risk into squared shrinkage bias and sampling variance. The objective of the empirical Bayes procedure is not to eliminate bias, but rather to reduce variance sufficiently to obtain a lower overall mean squared error.

To illustrate how network topology affects this trade-off, consider the special case

$$D = \tau^2 I$$

on the identified subspace and define

$$\eta = \frac{\sigma_\varepsilon^2}{\tau^2}.$$

Let

$$L_B = B'B.$$

Because the incidence matrix B contains two positive incidences per edge, L_B is the signless Laplacian associated with the bipartite graph. For a bipartite graph it has the same eigenvalues as the usual graph Laplacian after an appropriate sign transformation on one partition.

Let

$$0 = q_1 < q_2 \leq \dots \leq q_{N+M}$$

denote the eigenvalues of L_B under connectedness, with the zero eigenvalue corresponding to the location indeterminacy removed by the normalization. Let V denote the associated eigenvector matrix and define

$$b = V'(\alpha - m).$$

The conditional MSE of the empirical Bayes estimator becomes

$$\begin{aligned} \text{MSE}_{EB}(\alpha) &= \sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \\ &\quad + \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{q_r}{(q_r + \eta)^2}. \end{aligned} \tag{17}$$

The first component of (17) is squared shrinkage bias, whereas the second is variance.

For the normalized least-squares estimator,

$$\text{MSE}_{OLS} = \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{1}{q_r}.$$

It follows immediately that empirical Bayes has lower conditional MSE than least squares if and only if

$$\sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \leq \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \left[\frac{1}{q_r} - \frac{q_r}{(q_r + \eta)^2} \right]. \quad (18)$$

Condition (18) provides a direct economic interpretation of the gain from group-based shrinkage. The left-hand side is the cost of shrinking the individual effects toward their group means, whereas the right-hand side is the resulting reduction in estimation variance. If units are relatively homogeneous within groups, then $\alpha - m$ is small and the shrinkage bias is limited. At the same time, if the network contains weakly identified directions, corresponding to small positive eigenvalues q_r , the variance of the conventional two-way least-squares estimator becomes large. The potential benefit of empirical Bayes is therefore greatest when the latent groups are informative and the bipartite network is sparse or connected across groups.

The preceding comparison conditions on the realized heterogeneous effects. A stronger optimality result follows when the risk is integrated with respect to the correctly specified prior distribution. Under squared error loss, the posterior mean minimizes posterior expected loss. Consequently, for any alternative estimator $\delta(\tilde{Y})$,

$$R_B(\hat{\alpha}^{EB}) \leq R_B(\delta),$$

where

$$R_B(\delta) = \mathbb{E} \left[\left\| \delta(\tilde{Y}) - \alpha \right\|^2 \right]$$

integrates over both the sampling distribution and the group prior. In particular,

$$R_B(\hat{\alpha}^{EB}) \leq R_B(\hat{\alpha}^{OLS}).$$

In the scalar-prior case, the Bayes risk has an especially transparent spectral representation. Since

$$\eta = \frac{\sigma_\varepsilon^2}{\tau^2},$$

the posterior variance along spectral direction r is

$$\left(\frac{1}{\tau^2} + \frac{q_r}{\sigma_\varepsilon^2} \right)^{-1} = \frac{\sigma_\varepsilon^2}{q_r + \eta}.$$

Therefore,

$$R_B(\hat{\alpha}^{EB}) = \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{1}{q_r + \eta},$$

whereas

$$R_B(\hat{\alpha}^{OLS}) = \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{1}{q_r}.$$

Since $\eta > 0$ for a proper prior,

$$\frac{1}{q_r + \eta} < \frac{1}{q_r},$$

and hence

$$R_B(\hat{\alpha}^{EB}) < R_B(\hat{\alpha}^{OLS})$$

whenever the two estimators differ.

This result should be distinguished from Proposition ?? . For a fixed realization of α , empirical Bayes need not dominate least squares because excessive shrinkage toward an inaccurate group mean can increase squared bias. However, when risk is integrated over the correctly specified group prior, the posterior mean is optimal under quadratic loss by construction.

The spectral representation also illustrates why empirical Bayes is particularly useful for sparse network data. Consider a direction for which

$$q_r \rightarrow 0.$$

The least-squares contribution to risk is

$$\frac{\sigma_\varepsilon^2}{q_r},$$

which diverges as $q_r \rightarrow 0$. In contrast, the corresponding Bayes-risk contribution of the empirical Bayes estimator satisfies

$$\frac{\sigma_\varepsilon^2}{q_r + \eta} \rightarrow \frac{\sigma_\varepsilon^2}{\eta} = \tau^2.$$

The group prior therefore regularizes precisely those spectral directions that contain little information in the observed network.

This establishes a direct connection between network topology and the potential gain from

empirical Bayes estimation. Connectedness determines whether the normalized effects are identified, while the magnitude of the positive eigenvalues of the network Laplacian determines the strength of identification and hence the sampling variability of conventional two-way fixed-effect estimators. The proposed group-specific empirical Bayes estimator exploits the estimated latent structure to stabilize these weakly identified directions through partial pooling.

5 Simulated Study

In this section, we conduct a Monte Carlo simulation study to examine the finite-sample performance of the proposed machine-learning-assisted empirical Bayes (ML-EB) estimator. We consider a group prior of two-way unobserved heterogeneity with $K = L = 3$ latent groups on its two sides. The first side contains 9,000 units, with 3,000 units assigned to each group, and the second side contains 3,000 units, with 1,000 units assigned to each group. Let $g_i \in \{1, 2, 3\}$ and $h_j \in \{1, 2, 3\}$ denote the group memberships of node i on side s and node j on side t , respectively.

We generate the node-specific effects from group-specific Gaussian distributions:

$$\alpha_{s,i} \mid g_i = k \sim \mathcal{N}(\mu_k^s, (\tau_k^s)^2), \quad \alpha_{t,j} \mid h_j = \ell \sim \mathcal{N}(\mu_\ell^t, (\tau_\ell^t)^2).$$

The group-specific means are set to

$$\mu^s = (1.0, 0.5, 0.0)', \quad \mu^t = (0.6, 0.3, 0.0)',$$

and all group-specific prior variances are set to

$$\tau_1^s = \tau_2^s = \tau_3^s = \tau_1^t = \tau_2^t = \tau_3^t = 0.1.$$

Next, we define the one-hot membership vectors as

$$z_{s,i} = e_{g_i}, \quad z_{t,j} = e_{h_j},$$

where e_k denotes the k th canonical basis vector in $\mathbb{R}^3$. The group memberships Z are constructed

according to the realized values of α . For units on the s -side, we define

$$z_{s,i} = \begin{cases} [1, 0, 0], & \text{if } \alpha_{s,i} \in [0.75, 1.0], \\ [0, 1, 0], & \text{if } \alpha_{s,i} \in [0.0, 0.25], \\ [0, 0, 1], & \text{otherwise.} \end{cases}$$

Thus, units with relatively high values of $\alpha_{s,i}$ are assigned to the first group, those with relatively low values are assigned to the second group, and the remaining units are assigned to the third group. The same classification rule is applied to units on the t -side:

$$z_{t,j} = \begin{cases} [1, 0, 0], & \text{if } \alpha_{t,j} \in [0.45, 0.6], \\ [0, 1, 0], & \text{if } \alpha_{t,j} \in [0.0, 0.15], \\ [0, 0, 1], & \text{otherwise.} \end{cases}$$

Hence, the latent group memberships on both sides are determined by the locations of the corresponding heterogeneous effects within the support of α .

Conditional on the group memberships Z , we generate the bipartite network using a degree-corrected stochastic block model. For each pair (i, j) , the link indicator satisfies

$$A_{ij} \mid Z, \Omega, \Theta \sim \text{Bernoulli}(p_{ij}), \quad p_{ij} = \theta_{s,i} \theta_{t,j} z'_{s,i} \Omega z_{t,j},$$

independently across node pairs, where $\Theta = (\theta_{s,1}, \dots, \theta_{s,N}, \theta_{t,1}, \dots, \theta_{t,M})'$ collects the degree-correction parameters for all nodes on both sides. To isolate the role of group structure in the baseline design, we set

$$\theta_{s,i} = \theta_{t,j} = 1$$

for all i and j . The block-probability matrix is defined by

$$\Omega_{k\ell} = \begin{cases} 0.05, & k = \ell, \\ 0.001, & k \neq \ell. \end{cases}$$

Thus, links are substantially more likely to form between nodes belonging to groups with the same label than between nodes belonging to different groups.

For each observed link (i, j) , we independently generate one covariate and one idiosyncratic

disturbance according to

$$x_{ij} \sim \mathcal{N}(0, 1), \quad \varepsilon_{ij} \sim \mathcal{N}(0, 1),$$

and set the slope coefficient to $\beta = 0.3$. The edge-level outcome is then generated as

$$y_{ij} = \alpha_{s,i} + \alpha_{t,j} + x_{ij}\beta + \varepsilon_{ij}, \quad (i, j) \in \mathcal{E}.$$

After stacking the outcomes associated with all observed links, the data-generating process can be expressed as

$$Y = B\alpha + X\beta + \varepsilon,$$

where B is the edge-by-node design matrix constructed from the simulated bipartite network.

5.1 Main Result

We conduct 500 independent Monte Carlo replications. In each replication, we first apply the bi-spectral clustering procedure described above to the simulated bi-adjacency matrix A . The numbers of groups $\widehat{K}$ and $\widehat{L}$ in both sides are assumed to be known for the moment. This produces the estimated membership vectors

$$\widehat{z}_{s,i}, \quad i = 1, \dots, N,$$

for the nodes on side s , and

$$\widehat{z}_{t,j}, \quad j = 1, \dots, M,$$

for the nodes on side t . We then stack these vectors to obtain the estimated membership matrix $\widehat{Z}$. Next, we estimate β by OLS from the preliminary two-way fixed-effects regression and construct the residualized outcome

$$\widetilde{Y} = Y - X\widehat{\beta}.$$

Conditional on $\widehat{Z}$ and the edge-by-node design matrix B , we estimate the group-specific prior means μ_g and variances τ_g^2 by maximizing the marginal likelihood in Equation (11). Finally, we substitute B , $\widehat{Z}$, $\widehat{\mu}_g$, and $\widehat{\tau}_g^2$ into the posterior-mean expression in Equation (10) to obtain the feasible ML-EB estimator for that replication. The reported performance measures are calculated by averaging the corresponding results across the 500 replications.

We summarize the results in Table 1. The first panel of Table 1 reports the results for es-

timating the group-level parameters using marginal maximum likelihood. $\mu_{h_3} = 0$ is used as the identification condition. As shown in the first two columns, the true group means are set to $\mu_{g_1} = 1.0$, $\mu_{g_2} = 0.5$ and $\mu_{g_3} = 0$ on the first side, and $\mu_{h_1} = 0.6$, $\mu_{h_2} = 0.3$ and $\mu_{h_3} = 0$ on the second side. The fourth column reports the average estimated group means $\hat{\mu}$ across 500 simulation replications. The estimated values closely match the true parameter values used in the data-generating process. The fifth column reports the standard deviation of $\hat{\mu}$ across the 500 simulations, indicating that the estimators are both accurate and stable.

Table 1: Estimated group means and comparison between EB and OLS estimators.

			$ML - EB_{MLE}$	
			Mean	Std.
μ_{g_1}	1.0	$\hat{\mu}_{g_1}$	0.99	$1.32 * 10^{-4}$
μ_{g_2}	0.5	$\hat{\mu}_{g_2}$	0.498	$4.814 * 10^{-5}$
μ_{g_3}	0.0	$\hat{\mu}_{g_3}$	0.002	$1.706 * 10^{-4}$
μ_{h_1}	0.6	$\hat{\mu}_{h_1}$	0.599	$2.21 * 10^{-5}$
μ_{h_2}	0.3	$\hat{\mu}_{h_2}$	0.301	$3.36 * 10^{-4}$
μ_{h_3}	0.0			

OLS		$ML - EB_{MLE}$	
Mean	Median	Mean	Median
$RMSE$	0.339	0.031	0.031

Notes: first part reports the setup and estimated means in group prior using marginal MLE. The second part reports the root mean square error of mean and median for α over 500 simulation replications, using $\frac{1}{N+M} \sum_{i=1}^{N+M} \sqrt{(\alpha_i - \hat{\alpha}_i)^2}$. Moreover, μ_{h_3} is used as the identification condition.

The second panel of Table 1 presents the performance of the proposed machine-learning-assisted empirical Bayes (ML-EB) estimator. For comparison, we also report results from the ordinary least squares (OLS) estimator. Estimation accuracy is evaluated using the root mean squared error (RMSE), $\frac{1}{N+M} \sum_{i=1}^{N+M} \sqrt{(\alpha_i - \hat{\alpha}_i)^2}$. As shown in the final row of Table 1, the ML-EB estimator achieves a substantially lower average RMSE than the OLS estimator. This result indicates that group-specific shrinkage improves the recovery of the underlying heterogeneous effects under the data-generating processes considered. The second-panel results therefore demonstrate the effectiveness and finite-sample advantages of the proposed ML-EB estimator relative to unrestricted OLS.

6 Empirical Study

6.1 Amazon User Ratings

Online product ratings have become an important source of information for consumers, firms, and researchers studying digital markets. Existing research documents that Amazon ratings and reviews can affect product sales, consumers' acquisition of information, and consumer welfare (Chevalier and Mayzlin, 2006; Reimers and Waldfogel, 2021). Other studies emphasize that observed ratings may also be shaped by reviewer identity, selective reviewing, and the sequential evolution of consumer opinions (Godes and Silva, 2012; Forman et al., 2008). Accordingly, an observed rating may reflect not only the underlying appeal of a product but also the systematic rating tendency of the user providing the evaluation: some users consistently assign relatively high ratings, whereas others evaluate products more critically. Nevertheless, much of the existing literature relies on product-level average ratings or focuses on particular characteristics of reviews, leaving the two-sided heterogeneity underlying individual user-product evaluations relatively underexplored. In this subsection, we represent Amazon ratings as observations generated on a bipartite network connecting users and products and estimate a model incorporating both user- and product-specific heterogeneity,

$$\text{Rating} = \alpha_0 + B\alpha + \varepsilon,$$

where B is the incidence matrix derived from Amazon User-Product rating bipartite network. Furthermore, α_0 denotes Intercept and $\alpha = [\text{User}, \text{Product}]'$. The model decomposes each observed rating into a common component, a persistent user-specific rating tendency, a product-specific appeal component, and an idiosyncratic residual, thereby providing a more granular characterization of rating variation on both sides of the online marketplace.

We further allow the user-product bipartite network to exhibit latent group structure. In particular, users with similar interests may rate overlapping sets of products, while products with related characteristics may attract similar populations of users. For example, consumers interested in cooking may disproportionately rate kitchen products, students may concentrate on educational supplies and electronic devices, and sports enthusiasts may be more likely to rate sporting goods. These examples motivate, rather than impose, the presence of group structure; the latent groups are learned from the observed network using a data-driven procedure. Recov-

ering this structure allows us to estimate group-specific prior distributions of user and product effects through empirical Bayes shrinkage. This approach can improve estimation precision in sparse networks, characterize heterogeneity both within and across latent market segments, and potentially provide information relevant to marketing strategies directed toward particular consumer groups.

Our empirical analysis uses the Industrial and Scientific category of the Amazon Reviews’23 dataset compiled by the McAuley Lab at the University of California, San Diego. Amazon Reviews’23 is a large-scale collection of product reviews and user–product interactions spanning numerous product categories and an extended period. The complete Industrial and Scientific category contains approximately 5.2 million ratings submitted by 3.4 million users for approximately 427,500 products. Rather than using these raw data, we employ the officially provided 5-core, de-duplicated (*Pure IDs*) version. The 5-core procedure iteratively removes users and products with fewer than five distinct interactions, thereby retaining a denser portion of the bipartite network in which every included node has repeated connections. De-duplication removes duplicate review records and retains a single observation for each included user–product pair. These preprocessing steps substantially mitigate the weak-connectivity problem and limit the influence of users and products that appear only infrequently. This restriction is particularly useful in our setting because estimation of user- and product-specific heterogeneity requires sufficient overlap between the two sides of the network, while the precision of node-specific estimates generally deteriorates for low-degree nodes. The resulting estimation sample contains 412,947 distinct user–product rating observations involving 50,985 unique users and 25,848 unique parent products. The reviews in the estimation sample were posted between July 2001 and August 2023.

We represent the data as an undirected and unweighted bipartite network. As in Section 2, let $B \in \{0, 1\}^{n \times (N+M)}$ denote the edge-by-node incidence matrix used to define the two-way effect model, where n is the number of observed links and N and M are the numbers of users and products, respectively. For the network-clustering step in this section, it is more convenient to work with the corresponding bi-adjacency matrix $A \in \{0, 1\}^{N \times M}$, and

$$A_{ij} = \mathbf{1}\{\text{user } i \text{ rated product } j\}.$$

Writing $B = [B_s \ B_t]$, where $B_s \in \{0, 1\}^{n \times N}$ and $B_t \in \{0, 1\}^{n \times M}$ collect the user- and product-

side columns of B , respectively, the two representations are related by $A = B'_s B_t$: since the data are de-duplicated, each observed user–product pair contributes exactly one link, so every entry of $B'_s B_t$ is either 0 or 1 and coincides with A_{ij} . In our Amazon Industrial and Scientific sample, the resulting network contains $N = 50,985$ users, $M = 25,848$ products, and $n = 412,947$ observed user–product links.

We use the corrected BIC criterion to determine the number of groups (Hu et al., 2020),

$$\ell(K, L) = \max_{z, \theta} \log f(A \mid \theta, z) - [\lambda(N \log K + M \log L) + KL \log NM].$$

We then select

$$(\hat{K}, \hat{L}) = \arg \max_{K, L} \ell(K, L).$$

Applying the group-number selection procedure to the empirical network yields $\hat{K} = 10$ user groups and $\hat{L} = 11$ product groups. Given these selected numbers of groups, we construct spectral embeddings from the left and right singular vectors of the degree-normalized bi-adjacency matrix and apply clustering separately to the user and product embeddings (Rohe et al., 2016). Figure 3 presents the resulting group structure. Panel (a) displays the user–product bipartite network. Panel (b) displays the bi-adjacency matrix after its rows and columns have been reordered according to the estimated user and product memberships, respectively. The reordered matrix exhibits pronounced blockwise variation in link density, indicating that particular user groups interact disproportionately with particular product groups. Each rectangular block represents the interactions between one user group and one product group.

Next, given estimated group memberships, we estimate unobserved heterogeneity. The empirical Bayes procedure exploits the estimated group structure of the user–product network by allowing the distributions of the user and product effects to vary across their respective latent groups. We adopt group-specific Gaussian priors for both sets of effects. Because users are partitioned into 10 groups and products into 11 groups, the specification contains 21 group-specific prior distributions: 10 for the user effects and 11 for the product effects. This structure accommodates heterogeneity both among individual users and products within the same group and across network-defined groups through differences in the locations and dispersions of their prior distributions. Under the normalization imposed on the user and product effects, the intercept α_0 is estimated using preliminary FE method. The estimated intercept $\hat{\alpha}_0$ is 4.425. The detail of estimation is left to Section SM.3.3. Table 2 reports the estimated group-specific prior means

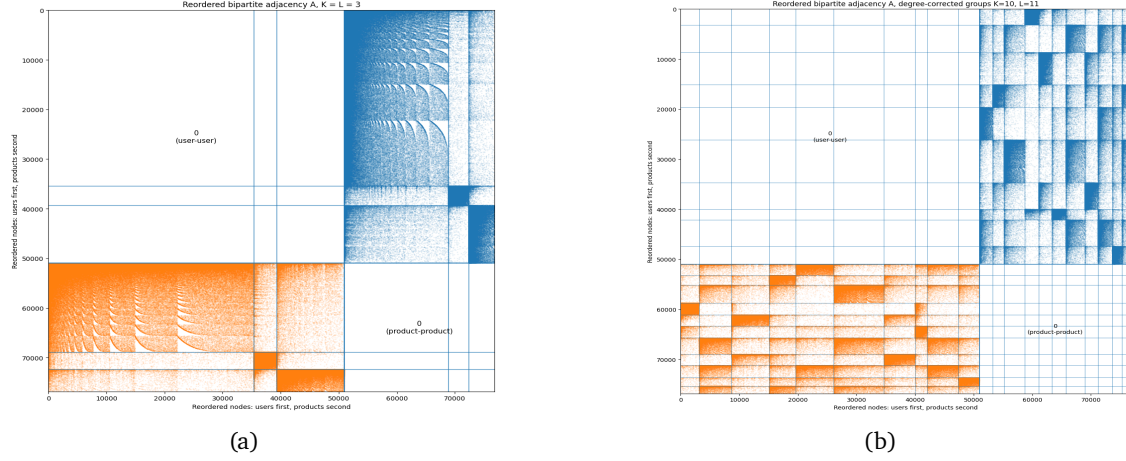


Figure 3: Panel (a) and (b) display the heatmaps of the corresponding bi-adjacency matrix, with both rows and columns reordered according to the estimated cluster memberships obtained from the bipartite spectral clustering procedure. In panel (a), both the numbers of groups in row and column are set to $K=L=3$. In panel (b), the number of groups in row is set to $K=10$ and the number of groups in column is set to $K=11$.

and standard deviations.

Table 2: Estimated group-specific prior parameters.

User Groups				Product Groups			
Group	N_{nodes}	Prior Mean	Prior SD	Group	N_{nodes}	Prior Mean	Prior SD
1	3,152	0.1444	0.1580	1	2,252	-0.0211	0.2651
2	5,531	0.0607	0.4068	2	1,903	0.0070	0.2434
3	6,429	-0.1709	0.6919	3	3,576	0.0642	0.2187
4	4,515	0.0353	0.4529	4	2,387	0.1207	0.0945
5	6,525	0.0152	0.4921	5	2,241	-0.1756	0.3819
6	8,558	0.0621	0.4136	6	2,375	-0.0119	0.2241
7	5,295	-0.0746	0.5927	7	3,270	0.0737	0.1997
8	2,085	0.0998	0.2980	8	2,188	-0.1789	0.3864
9	5,314	0.0321	0.4515	9	2,473	0.0081	0.2520
10	3,581	0.0120	0.4753	10	1,655	0.0163	0.2615
				11	1,528	0.0273	0.2334

Notes: Group means and variances are estimated jointly by marginal maximum likelihood, conditional on the supplied group memberships. The FE intercept and innovation variance are held fixed at $\hat{\beta}_0 = 4.425580$ and $\hat{\sigma}_\varepsilon^2 = 0.846414$. Prior SD denotes the square root of the estimated prior variance. Product-group means satisfy the normalization $\sum_{\ell=1}^{11} (M_\ell/M) \hat{\mu}_\ell^T = 0$, where M_ℓ is the number of products in group ℓ .

The estimates reveal substantial heterogeneity in rating behavior across user groups. In particular, the estimated prior means of the user effects range from -0.1709 for User Group 3 to 0.1444 for User Group 1. Because the dependent variable is measured in rating points, these estimates have a direct interpretation under the adopted normalization. Holding the product-specific effect constant, a user in Group 1 assigns ratings that are, on average, approximately 0.1444

points above the common benchmark, whereas a user in Group 3 assigns ratings approximately 0.1709 points below it. The difference between the two group-specific prior means is therefore 0.3154 rating points. This result suggests that the groups inferred exclusively from the interaction network also capture substantively meaningful differences in rating behavior. One possible interpretation is that Group 1 contains relatively favorable or generous reviewers, whereas Group 3 contains comparatively demanding or critical reviewers. Because the model estimates product-specific effects simultaneously, these differences are not mechanically attributable to some user groups reviewing products with systematically higher or lower average ratings. Nevertheless, the estimates describe conditional rating tendencies and should not be interpreted as causal effects of group membership.

The estimated group-specific prior variances reveal an additional dimension of heterogeneity in user rating behavior. User Group 1 has an estimated prior variance of only 0.0250, corresponding to a prior standard deviation of approximately 0.1580 rating points, whereas User Group 3 has an estimated prior variance of 0.4787, corresponding to a standard deviation of approximately 0.6919. Thus, Group 1 is characterized not only by a relatively high prior mean but also by comparatively homogeneous user effects. In contrast, the effects of users in Group 3 are much more widely dispersed around the group's negative prior mean, indicating substantial within-group variation in rating tendencies. This contrast illustrates an important advantage of the group-specific empirical Bayes approach. Imposing a common prior distribution on all users would obscure both differences in average rating tendencies across groups and differences in within-group dispersion. By estimating a separate prior distribution for each group, the procedure allows the shrinkage pattern to adapt to group-specific heterogeneity. Holding the information available for an individual effect fixed, a smaller prior variance produces stronger shrinkage toward the corresponding group mean, whereas a larger prior variance preserves more individual-level variation. In the joint two-way effect model, the actual amount of posterior shrinkage also depends on the number and pattern of observed ratings and on the dependence induced by the bipartite network design.

A similarly pronounced pattern emerges for the product effects. The estimated product-group prior mean is lowest for Product Group 8 at approximately -0.1789 , followed closely by Product Group 5 at -0.1756 , and is highest for Product Group 4 at approximately 0.1207 . Under the adopted normalization and holding user-specific rating tendencies constant, products in Group 4 have a systematically higher average product-specific rating component than products in Groups

5 and 8. In particular, the difference between the prior means of Product Groups 4 and 8 is 0.2996 rating points. The estimated group structure may capture similarities in product characteristics, market segments, user–product matching patterns, or other latent attributes associated with product ratings. Importantly, the groups are estimated exclusively from the binary interaction structure rather than from the numerical ratings or observed product characteristics. The association between these network-defined groups and the estimated product effects therefore indicates that reviewing patterns contain useful information about product-side rating heterogeneity. Nevertheless, the product effects should be interpreted as conditional rating components rather than as causal measures of product quality.

The estimated product-group prior variances provide an additional dimension of product-side heterogeneity. Product Group 4 has an estimated prior variance of only 0.0089, corresponding to a prior standard deviation of approximately 0.0945 rating points. This dispersion is substantially smaller than the estimated variances of 0.1459 and 0.1493 for Product Groups 5 and 8, which correspond to standard deviations of approximately 0.3819 and 0.3864, respectively. The relatively small prior variance for Group 4 indicates that the product effects in this group are tightly concentrated around their group-specific mean, making that mean particularly informative about an individual product’s latent effect. Holding product-level information and network position fixed, the empirical Bayes estimator therefore shrinks the effects of products in Group 4 more strongly toward their group mean. By contrast, the larger prior variances for Groups 5 and 8 indicate substantial product-level heterogeneity even after conditioning on group membership, leading the estimator to preserve more of the individual variation in product effects. These findings are consistent with a flexible hierarchical representation in which network-defined groups provide useful information about product similarity without imposing identical effects on all products within the same group.

Empirical Bayes shrinkage is particularly important in the present application because node degrees vary considerably across users and products, even within the 5-core sample. An unrestricted two-way fixed-effects estimator treats every user and product effect as a separate parameter and imposes no distributional pooling across nodes. Consequently, effects associated with relatively few or weakly informative network connections may be estimated with considerable noise. The empirical Bayes estimator addresses this problem by combining the information contained in each node’s observed ratings with the estimated prior distribution of its assigned group. All else being equal, an imprecisely estimated individual effect is shrunk more strongly toward

its group-specific prior mean, whereas a precisely estimated effect receives greater weight from its individual data and is shrunk less. The amount of shrinkage depends not only on the number of observed ratings but also on the configuration of network links and the estimated variance of the relevant group-specific prior. Crucially, shrinkage is not directed toward a single population-wide mean. A user assigned to User Group k is shrunk toward the estimated prior mean of that user group, whereas a product assigned to Product Group ℓ is shrunk toward the corresponding product-group mean. Thus, prior-induced information sharing occurs primarily among nodes with similar network interaction patterns rather than through indiscriminate pooling across all users or products. At the same time, the joint posterior continues to account for the dependence between user and product effects induced by the observed bipartite network.

Taken together, the results indicate that the network-based group structure contains substantively relevant information about heterogeneity on both sides of the Amazon user–product rating network. Although numerical ratings are not used to construct the groups, users assigned to different network-defined groups exhibit systematically different rating tendencies and degrees of within-group dispersion. Similarly, product groups differ in both their average product-specific rating components and their within-group heterogeneity.

7 Conclusion

We study the group structure in two-way effect model driven by a bipartite network induced by matched data. The model is organized using a directed acyclic graph that not only explicitly characterizes the latent data-generating mechanism but also leads to a new framework for empirical Bayes estimation. Building on this DAG, we develop a specific machine-learning-assisted empirical Bayes estimator, motivated by the poor performance of ordinary least squares in networks that are sparse and exhibit latent group structure. We establish identification and derive theoretical results characterizing the properties of the proposed estimator.

In addition, we introduce a novel group prior that allows for within-group heterogeneity while preserving latent group structure. An application of the proposed methodology to the Amazon user–product ratings data reveals substantial group-structured heterogeneity in user and product effects, which can help inform more precise and targeted marketing strategies directed at different groups of users.

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Appendix A. Proof

Proof of Proposition 1. Since

$$B(\alpha_1 - \alpha_2) = 0,$$

Assumption 2 implies

$$\alpha_1 - \alpha_2 = cv_0$$

for some $c \in \mathbb{R}$. Because both parameter vectors satisfy the same normalization, Assumption 1 implies $c = 0$. Therefore $\alpha_1 = \alpha_2$. $\square$

Proof of Proposition 2. Let A denote the observed bipartite adjacency matrix and let

$$\mathcal{A} = \mathbb{E}[A \mid Z, \theta]$$

denote its population counterpart. Under the degree-corrected stochastic block model,

$$\mathcal{A} = \Theta_s Z_s P Z_t' \Theta_t,$$

where Θ_s and Θ_t collect the degree parameters on the two sides of the network.

By Assumption 6, the group proportions are bounded away from zero and the rows and columns of P corresponding to different latent groups are separated. Consequently, the population left and right singular-vector embeddings of $\mathcal{A}$ contain K and L distinct population centers, respectively, one for each latent group.

The minimum-degree condition implies concentration of the sample adjacency matrix around its population counterpart. In particular,

$$\|A - \mathcal{A}\|_{\text{op}} = o_p\left(\sigma_{\min}^+(\mathcal{A})\right),$$

where $\sigma_{\min}^+(\mathcal{A})$ denotes the smallest nonzero singular value relevant for recovering the latent subspaces.

Standard singular-subspace perturbation results therefore imply that, after an appropriate orthogonal transformation, the estimated left and right singular-vector matrices converge to

their population counterparts:

$$\|\widehat{U} - UO_s\|_F = o_p(\sqrt{N}), \quad \|\widehat{V} - VO_t\|_F = o_p(\sqrt{M}).$$

Because the population group centers are separated by strictly positive constants, applying K -means to $\widehat{U}$ and L -means to $\widehat{V}$ yields vanishing misclassification proportions. Hence there exist permutation matrices Π_s and Π_t such that

$$\frac{1}{N} \|\widehat{Z}_s - Z_s \Pi_s\|_0 \xrightarrow{p} 0$$

and

$$\frac{1}{M} \|\widehat{Z}_t - Z_t \Pi_t\|_0 \xrightarrow{p} 0.$$

This proves the result. $\square$

Proof of Proposition 3. Define the normalized feasible and infeasible marginal likelihood criteria by

$$Q_n(\psi, \widehat{Z}) = \frac{1}{a_n} \ell_n(\psi; \widehat{Z}), \tag{19}$$

$$Q_n(\psi, Z) = \frac{1}{a_n} \ell_n(\psi; Z). \tag{20}$$

Recall that the feasible empirical Bayes hyperparameter estimator is

$$\widehat{\psi} = \arg \max_{\psi \in \Psi} Q_n(\psi, \widehat{Z}). \tag{21}$$

We prove that

$$\widehat{\psi} \xrightarrow{p} \psi_0.$$

The proof consists of three steps.

Step 1: Uniform convergence of the feasible criterion.

By the triangle inequality,

$$\begin{aligned} \sup_{\psi \in \Psi} |Q_n(\psi, \widehat{Z}) - Q(\psi)| &\leq \sup_{\psi \in \Psi} |Q_n(\psi, \widehat{Z}) - Q_n(\psi, Z)| \\ &\quad + \sup_{\psi \in \Psi} |Q_n(\psi, Z) - Q(\psi)|. \end{aligned} \tag{22}$$

By condition (15),

$$\sup_{\psi \in \Psi} |Q_n(\psi, \hat{Z}) - Q_n(\psi, Z)| = o_p(1). \quad (23)$$

Moreover, Assumption 9 implies

$$\sup_{\psi \in \Psi} |Q_n(\psi, Z) - Q(\psi)| = o_p(1). \quad (24)$$

Combining (22)–(24) therefore gives

$$\sup_{\psi \in \Psi} |Q_n(\psi, \hat{Z}) - Q(\psi)| = o_p(1). \quad (25)$$

Hence the feasible marginal likelihood criterion converges uniformly in probability to the same population criterion $Q(\psi)$ as the infeasible criterion based on the true group membership Z .

Under exact group recovery, condition (23) follows immediately. Indeed, after an appropriate permutation of group labels,

$$\Pr(\hat{Z} = Z) \rightarrow 1.$$

On the event $\{\hat{Z} = Z\}$,

$$Q_n(\psi, \hat{Z}) = Q_n(\psi, Z)$$

for every $\psi \in \Psi$. Therefore,

$$\sup_{\psi \in \Psi} |Q_n(\psi, \hat{Z}) - Q_n(\psi, Z)| = 0$$

with probability approaching one, which implies (23).

Step 2: Separation of the population objective.

Fix an arbitrary $\epsilon > 0$. Define

$$\Psi_\epsilon = \{\psi \in \Psi : \|\psi - \psi_0\| \geq \epsilon\}. \quad (26)$$

Because Ψ is compact, Ψ_ϵ is also compact. By Assumption 7, ψ_0 is the unique identified true parameter, and by Assumption 9, $Q(\psi)$ is uniquely maximized at ψ_0 .

Assume in addition that $Q(\psi)$ is continuous on Ψ . Since Ψ_ϵ is compact, the maximum of Q

over Ψ_ϵ exists. Uniqueness of ψ_0 implies

$$\sup_{\psi \in \Psi_\epsilon} Q(\psi)Q(\psi_0). \quad (27)$$

Define

$$\Delta_\epsilon = Q(\psi_0) - \sup_{\psi \in \Psi_\epsilon} Q(\psi). \quad (28)$$

Then

$$\Delta_\epsilon > 0. \quad (29)$$

This establishes a strictly positive population criterion gap between the true parameter ψ_0 and every parameter lying at least ϵ away from ψ_0 .

Step 3: Consistency of the argmax.

Let

$$\mathcal{A}_n(\epsilon) = \left\{ \sup_{\psi \in \Psi} |Q_n(\psi, \hat{Z}) - Q(\psi)| < \frac{\Delta_\epsilon}{3} \right\}. \quad (30)$$

By (25),

$$\Pr(\mathcal{A}_n(\epsilon)) \rightarrow 1. \quad (31)$$

On the event $\mathcal{A}_n(\epsilon)$, evaluation of the feasible criterion at the true parameter gives

$$Q_n(\psi_0, \hat{Z}) > Q(\psi_0) - \frac{\Delta_\epsilon}{3}. \quad (32)$$

For any $\psi \in \Psi_\epsilon$, again on $\mathcal{A}_n(\epsilon)$,

$$\begin{aligned} & Q_n(\psi, \hat{Z})Q(\psi) + \frac{\Delta_\epsilon}{3} \\ & \leq \sup_{\tilde{\psi} \in \Psi_\epsilon} Q(\tilde{\psi}) + \frac{\Delta_\epsilon}{3}. \end{aligned} \quad (33)$$

Using the definition of Δ_ϵ in (28),

$$\begin{aligned} & \sup_{\psi \in \Psi_\epsilon} Q_n(\psi, \hat{Z})Q(\psi_0) - \Delta_\epsilon + \frac{\Delta_\epsilon}{3} \\ & = Q(\psi_0) - \frac{2\Delta_\epsilon}{3}. \end{aligned} \quad (34)$$

On the other hand, (32) gives

$$Q_n(\psi_0, \hat{Z}) > Q(\psi_0) - \frac{\Delta_\epsilon}{3}. \quad (35)$$

Since

$$Q(\psi_0) - \frac{\Delta_\epsilon}{3} > Q(\psi_0) - \frac{2\Delta_\epsilon}{3},$$

we obtain

$$Q_n(\psi_0, \hat{Z}) > \sup_{\psi \in \Psi_\epsilon} Q_n(\psi, \hat{Z}) \quad (36)$$

on $\mathcal{A}_n(\epsilon)$.

Therefore, on $\mathcal{A}_n(\epsilon)$, a maximizer of the feasible criterion cannot belong to Ψ_ϵ . By (21),

$$\hat{\psi} \notin \Psi_\epsilon. \quad (37)$$

Equivalently,

$$\|\hat{\psi} - \psi_0\| \geq \epsilon. \quad (38)$$

It follows that

$$\begin{aligned} \Pr \left(\|\hat{\psi} - \psi_0\| \geq \epsilon \right) &\leq \Pr \left(\mathcal{A}_n(\epsilon)^c \right) \\ &\longrightarrow 0. \end{aligned} \quad (39)$$

Since $\epsilon > 0$ was arbitrary,

$$\hat{\psi} \xrightarrow{p} \psi_0. \quad (40)$$

Finally, recall that

$$\psi = \left(\mu', (\tau^2)', \sigma_\epsilon^2 \right)'.$$

Convergence of the finite-dimensional vector $\hat{\psi}$ therefore implies componentwise convergence.

Hence, after aligning the estimated group labels with their population labels,

$$\hat{\mu}_k^s \xrightarrow{p} \mu_k^s, \quad \hat{\tau}_{k,s}^2 \xrightarrow{p} \tau_{k,s}^2, \quad k = 1, \dots, K, \quad (41)$$

$$\hat{\mu}_\ell^t \xrightarrow{p} \mu_\ell^t, \quad \hat{\tau}_{\ell,t}^2 \xrightarrow{p} \tau_{\ell,t}^2, \quad \ell = 1, \dots, L, \quad (42)$$

and

$$\hat{\sigma}_\epsilon^2 \xrightarrow{p} \sigma_\epsilon^2. \quad (43)$$

This establishes the stated consistency result. $\square$

Proof of Theorem 1. Throughout the proof, group labels are aligned with their population counterparts. This is without loss of generality because latent-group labels are identified only up to permutation.

Let

$$p_n = N + M,$$

and define

$$m = Z\mu, \quad \hat{m} = \hat{Z}\hat{\mu},$$

as well as

$$D = D(\tau^2), \quad \hat{D} = D(\hat{\tau}^2).$$

The oracle empirical Bayes estimator is

$$\alpha^{EB,O} = m + DB' \left(BDB' + \sigma_\varepsilon^2 I_n \right)^{-1} \left(\tilde{Y} - Bm \right), \quad (44)$$

whereas the feasible estimator is

$$\hat{\alpha}^{EB} = \hat{m} + \hat{D}B' \left(B\hat{D}B' + \hat{\sigma}_\varepsilon^2 I_n \right)^{-1} \left(\tilde{Y} - B\hat{m} \right). \quad (45)$$

We establish the result in five steps.

Step 1: Convergence of the estimated prior mean vector.

By exact group recovery,

$$\Pr(\hat{Z} = Z) \rightarrow 1$$

after aligning the group labels. Hence, with probability approaching one,

$$\hat{m} - m = Z(\hat{\mu} - \mu).$$

Since every row of Z contains exactly one nonzero element,

$$\begin{aligned} \frac{1}{p_n} \|\hat{m} - m\|^2 &= \frac{1}{p_n} \sum_{r=1}^{K+L} n_r (\hat{\mu}_r - \mu_r)^2 \\ &\leq \max_{1 \leq r \leq K+L} |\hat{\mu}_r - \mu_r|^2, \end{aligned} \quad (46)$$

where n_r denotes the number of nodes belonging to group r .

Because $K + L$ is fixed and Proposition 3 implies

$$\hat{\mu}_r \xrightarrow{p} \mu_r$$

for every r ,

$$\max_{1 \leq r \leq K+L} |\hat{\mu}_r - \mu_r| = o_p(1). \quad (47)$$

Therefore,

$$\frac{1}{p_n} \|\hat{m} - m\|^2 = o_p(1). \quad (48)$$

Step 2: Convergence of the prior precision matrices.

Define

$$\Lambda = \sigma_\varepsilon^2 D^{-1} \quad (49)$$

and

$$\hat{\Lambda} = \hat{\sigma}_\varepsilon^2 \hat{D}^{-1}. \quad (50)$$

Under exact group recovery, both D and $\hat{D}$ are diagonal matrices whose diagonal elements correspond to the same latent groups. Consequently,

$$\|\hat{D} - D\|_{\text{op}} = \max_{1 \leq r \leq K+L} |\hat{\tau}_r^2 - \tau_r^2|. \quad (51)$$

By Proposition 3,

$$\hat{\tau}_r^2 \xrightarrow{p} \tau_r^2$$

for every r . Since the number of groups is fixed,

$$\|\hat{D} - D\|_{\text{op}} = o_p(1). \quad (52)$$

The prior variances are bounded away from zero by assumption. Therefore matrix inversion is continuous in a neighborhood of D , and

$$\|\hat{D}^{-1} - D^{-1}\|_{\text{op}} = o_p(1). \quad (53)$$

Moreover,

$$\hat{\sigma}_\varepsilon^2 \xrightarrow{p} \sigma_\varepsilon^2.$$

Hence

$$\begin{aligned}
\|\hat{\Lambda} - \Lambda\|_{\text{op}} &= \left\| \hat{\sigma}_\varepsilon^2 \hat{D}^{-1} - \sigma_\varepsilon^2 D^{-1} \right\|_{\text{op}} \\
&\leq \left| \hat{\sigma}_\varepsilon^2 - \sigma_\varepsilon^2 \right| \|\hat{D}^{-1}\|_{\text{op}} \\
&\quad + \sigma_\varepsilon^2 \|\hat{D}^{-1} - D^{-1}\|_{\text{op}} \\
&= o_p(1).
\end{aligned} \tag{54}$$

Step 3: Stability of the empirical Bayes updating matrix.

Using the standard matrix identity

$$DB' \left(BDB' + \sigma_\varepsilon^2 I_n \right)^{-1} = \left(B'B + \sigma_\varepsilon^2 D^{-1} \right)^{-1} B', \tag{55}$$

define

$$A = (B'B + \Lambda)^{-1} B', \tag{56}$$

$$\hat{A} = (B'B + \hat{\Lambda})^{-1} B'. \tag{57}$$

Let

$$H = B'B + \Lambda, \quad \hat{H} = B'B + \hat{\Lambda}.$$

Since $B'B$ is positive semidefinite,

$$\lambda_{\min}(H) \geq \lambda_{\min}(\Lambda).$$

By the uniform lower and upper bounds imposed on τ_r^2 and σ_ε^2 , there exists a constant $c_\Lambda > 0$ such that

$$\lambda_{\min}(\Lambda) \geq c_\Lambda. \tag{58}$$

Therefore

$$\|H^{-1}\|_{\text{op}} \leq c_\Lambda^{-1}. \tag{59}$$

By (54), with probability approaching one,

$$\lambda_{\min}(\hat{\Lambda}) \geq \frac{c_\Lambda}{2},$$

so

$$\|\hat{H}^{-1}\|_{\text{op}} = O_p(1). \quad (60)$$

Using the resolvent identity,

$$\hat{H}^{-1} - H^{-1} = \hat{H}^{-1} (\Lambda - \hat{\Lambda}) H^{-1}. \quad (61)$$

It follows that

$$\begin{aligned} \hat{A} - A &= (\hat{H}^{-1} - H^{-1}) B' \\ &= \hat{H}^{-1} (\Lambda - \hat{\Lambda}) H^{-1} B' \\ &= \hat{H}^{-1} (\Lambda - \hat{\Lambda}) A. \end{aligned} \quad (62)$$

We next show that $\|A\|_{\text{op}}$ is bounded. Observe that

$$H = B'B + \Lambda \succeq B'B.$$

Therefore

$$H^{-1/2} B' B H^{-1/2} \preceq I_{p_n}. \quad (63)$$

Consequently,

$$\begin{aligned} \|A\|_{\text{op}} &= \|H^{-1} B'\|_{\text{op}} \\ &\leq \|H^{-1/2}\|_{\text{op}} \|H^{-1/2} B'\|_{\text{op}} \\ &\leq c_{\Lambda}^{-1/2}. \end{aligned} \quad (64)$$

Combining (54), (60), (62), and (64), we obtain

$$\|\hat{A} - A\|_{\text{op}} = o_p(1). \quad (65)$$

We will also require convergence of the corresponding shrinkage operators. Since

$$AB = H^{-1} B' B = I - H^{-1} \Lambda,$$

and similarly

$$\hat{A}B = I - \hat{H}^{-1}\hat{\Lambda},$$

we have

$$\begin{aligned}\hat{A}B - AB &= H^{-1}\Lambda - \hat{H}^{-1}\hat{\Lambda} \\ &= (H^{-1} - \hat{H}^{-1})\Lambda + \hat{H}^{-1}(\Lambda - \hat{\Lambda}).\end{aligned}\tag{66}$$

Using (61) and the boundedness established above,

$$\|\hat{A}B - AB\|_{\text{op}} = o_p(1).\tag{67}$$

Moreover,

$$I - \hat{A}B = \hat{H}^{-1}\hat{\Lambda},\tag{68}$$

so

$$\|I - \hat{A}B\|_{\text{op}} = O_p(1).\tag{69}$$

Step 4: Decomposition of the feasible-oracle difference.

Using the compact representations

$$\hat{\alpha}^{EB} = \hat{m} + \hat{A}(\tilde{Y} - B\hat{m}),\tag{70}$$

$$\alpha^{EB,O} = m + A(\tilde{Y} - Bm),\tag{71}$$

we obtain

$$\begin{aligned}\hat{\alpha}^{EB} - \alpha^{EB,O} &= \hat{m} - m - \hat{A}B(\hat{m} - m) \\ &\quad + (\hat{A} - A)(\tilde{Y} - Bm) \\ &= (I - \hat{A}B)(\hat{m} - m) \\ &\quad + (\hat{A} - A)(\tilde{Y} - Bm).\end{aligned}\tag{72}$$

Since

$$\tilde{Y} = B\alpha + \varepsilon,$$

we have

$$\tilde{Y} - Bm = B(\alpha - m) + \varepsilon.$$

Hence

$$\begin{aligned}\hat{\alpha}^{EB} - \alpha^{EB,O} &= (I - \hat{A}B)(\hat{m} - m) \\ &\quad + (\hat{A} - A)B(\alpha - m) \\ &\quad + (\hat{A} - A)\varepsilon.\end{aligned}\tag{73}$$

We treat the three terms separately.

Step 5: Each component is asymptotically negligible.

For the first term, using (69) and (48),

$$\begin{aligned}\frac{1}{p_n} \left\| (I - \hat{A}B)(\hat{m} - m) \right\|^2 &\leq \|I - \hat{A}B\|_{\text{op}}^2 \frac{\|\hat{m} - m\|^2}{p_n} \\ &= O_p(1) o_p(1) \\ &= o_p(1).\end{aligned}\tag{74}$$

For the second term, from (67),

$$\begin{aligned}\frac{1}{p_n} \left\| (\hat{A} - A)B(\alpha - m) \right\|^2 &= \frac{1}{p_n} \left\| (\hat{A}B - AB)(\alpha - m) \right\|^2 \\ &\leq \|\hat{A}B - AB\|_{\text{op}}^2 \frac{\|\alpha - m\|^2}{p_n}.\end{aligned}\tag{75}$$

By assumption,

$$\frac{1}{p_n} \|\alpha - m\|^2 = O_p(1),$$

while

$$\|\hat{A}B - AB\|_{\text{op}} = o_p(1).$$

Therefore

$$\frac{1}{p_n} \left\| (\hat{A} - A)B(\alpha - m) \right\|^2 = o_p(1).\tag{76}$$

For the third term, using (62),

$$(\hat{A} - A)\varepsilon = \hat{H}^{-1}(\Lambda - \hat{\Lambda})A\varepsilon.\tag{77}$$

Hence

$$\frac{1}{p_n} \|(\hat{A} - A)\varepsilon\|^2 \leq \|\hat{H}^{-1}\|_{\text{op}}^2 \|\hat{\Lambda} - \Lambda\|_{\text{op}}^2 \frac{\|A\varepsilon\|^2}{p_n}. \quad (78)$$

It remains to show that

$$\frac{1}{p_n} \|A\varepsilon\|^2 = O_p(1).$$

Conditional on the network B ,

$$\begin{aligned} \mathbb{E} \left[\frac{1}{p_n} \|A\varepsilon\|^2 \mid B \right] &= \frac{\sigma_\varepsilon^2}{p_n} \text{tr}(AA') \\ &\leq \sigma_\varepsilon^2 \|A\|_{\text{op}}^2 \\ &\leq \sigma_\varepsilon^2 c_\Lambda^{-1}. \end{aligned} \quad (79)$$

The final expression is uniformly bounded. By Markov's inequality,

$$\frac{1}{p_n} \|A\varepsilon\|^2 = O_p(1). \quad (80)$$

Combining (60), (54), and (80) in (78) yields

$$\frac{1}{p_n} \|(\hat{A} - A)\varepsilon\|^2 = o_p(1). \quad (81)$$

Finally, using

$$\|a + b + c\|^2 \leq 3\|a\|^2 + 3\|b\|^2 + 3\|c\|^2$$

together with (74), (76), and (81), we obtain

$$\frac{1}{p_n} \left\| \hat{\alpha}^{EB} - \alpha^{EB,O} \right\|^2 = o_p(1). \quad (82)$$

Since $p_n = N + M$,

$$\frac{1}{N + M} \left\| \hat{\alpha}^{EB} - \alpha^{EB,O} \right\|^2 \xrightarrow{p} 0. \quad (83)$$

This proves the proposition. $\square$

Proof of Proposition 4. All calculations below are carried out on the normalized identified subspace, so that the zero eigenvalue associated with the location indeterminacy is excluded.

Under the scalar-prior specification

$$D = \tau^2 I,$$

the empirical Bayes posterior mean can be equivalently written as

$$\hat{\alpha}^{EB} = m + (Q_B + \eta I)^{-1} B' (\tilde{Y} - Bm), \quad (84)$$

where

$$Q_B = B' B, \quad \eta = \frac{\sigma_\varepsilon^2}{\tau^2}.$$

Using

$$\tilde{Y} = B\alpha + \varepsilon,$$

we obtain

$$\begin{aligned} \hat{\alpha}^{EB} - \alpha &= m - \alpha + (Q_B + \eta I)^{-1} B' [B(\alpha - m) + \varepsilon] \\ &= m - \alpha + (Q_B + \eta I)^{-1} Q_B (\alpha - m) + (Q_B + \eta I)^{-1} B' \varepsilon. \end{aligned} \quad (85)$$

Since

$$(Q_B + \eta I)^{-1} Q_B = I - \eta (Q_B + \eta I)^{-1},$$

equation (85) reduces to

$$\hat{\alpha}^{EB} - \alpha = -\eta (Q_B + \eta I)^{-1} (\alpha - m) + (Q_B + \eta I)^{-1} B' \varepsilon. \quad (86)$$

The first term in (86) is the shrinkage-bias component and the second term is the sampling-error component.

Let the eigendecomposition of Q_B be

$$Q_B = V Q V', \quad (87)$$

where

$$Q = \text{diag}(q_1, \dots, q_{N+M})$$

and $V'V = I$. Define

$$b = V'(\alpha - m).$$

Conditional on the realized heterogeneous effects α , the error ε has mean zero. Hence

$$\mathbb{E} [\hat{\alpha}^{EB} - \alpha \mid \alpha] = -\eta(Q_B + \eta I)^{-1}(\alpha - m). \quad (88)$$

The squared conditional bias is therefore

$$\begin{aligned} \left\| \mathbb{E} [\hat{\alpha}^{EB} - \alpha \mid \alpha] \right\|^2 &= \eta^2(\alpha - m)'(Q_B + \eta I)^{-2}(\alpha - m) \\ &= \eta^2 b'(Q + \eta I)^{-2} b. \end{aligned} \quad (89)$$

Restricting attention to the identified subspace gives

$$\text{Bias}_{EB}^2(\alpha) = \sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2}. \quad (90)$$

Next consider the variance term. Conditional on α ,

$$\text{Var}(\hat{\alpha}^{EB} \mid \alpha) = \sigma_\varepsilon^2 (Q_B + \eta I)^{-1} B' B (Q_B + \eta I)^{-1}. \quad (91)$$

Since $B' B = Q_B$,

$$\text{Var}(\hat{\alpha}^{EB} \mid \alpha) = \sigma_\varepsilon^2 (Q_B + \eta I)^{-1} Q_B (Q_B + \eta I)^{-1}. \quad (92)$$

Taking the trace and using the spectral decomposition yields

$$\begin{aligned} \text{tr} [\text{Var}(\hat{\alpha}^{EB} \mid \alpha)] &= \sigma_\varepsilon^2 \text{tr} [(Q + \eta I)^{-1} Q (Q + \eta I)^{-1}] \\ &= \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{q_r}{(q_r + \eta)^2}. \end{aligned} \quad (93)$$

Combining (90) and (93), the conditional mean squared error of the empirical Bayes estimator is

$$\begin{aligned} \text{MSE}_{EB}(\alpha) &= \sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \\ &\quad + \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{q_r}{(q_r + \eta)^2}. \end{aligned} \quad (94)$$

We next derive the corresponding MSE for the normalized least-squares estimator. On the identified subspace,

$$\hat{\alpha}^{OLS} = Q_B^+ B' \tilde{Y}, \quad (95)$$

where Q_B^+ denotes the Moore–Penrose inverse of Q_B , restricted to the normalized parameter space. Because

$$\tilde{Y} = B\alpha + \varepsilon,$$

and

$$Q_B^+ Q_B \alpha = \alpha$$

on this subspace,

$$\hat{\alpha}^{OLS} - \alpha = Q_B^+ B' \varepsilon. \quad (96)$$

Therefore OLS is conditionally unbiased:

$$\mathbb{E} [\hat{\alpha}^{OLS} - \alpha \mid \alpha] = 0. \quad (97)$$

Its conditional covariance matrix is

$$\begin{aligned} \text{Var}(\hat{\alpha}^{OLS} \mid \alpha) &= \sigma_\varepsilon^2 Q_B^+ B' B Q_B^+ \\ &= \sigma_\varepsilon^2 Q_B^+. \end{aligned} \quad (98)$$

Consequently,

$$\text{MSE}_{OLS} = \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{1}{q_r}. \quad (99)$$

We can now compare the two estimators. From (94) and (99),

$$\begin{aligned} &\text{MSE}_{OLS} - \text{MSE}_{EB}(\alpha) \\ &= \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{1}{q_r} - \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{q_r}{(q_r + \eta)^2} - \sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \\ &= \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \left[\frac{1}{q_r} - \frac{q_r}{(q_r + \eta)^2} \right] - \sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2}. \end{aligned} \quad (100)$$

Hence

$$\text{MSE}_{EB}(\alpha) \leq \text{MSE}_{OLS}$$

if and only if

$$\sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \leq \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \left[\frac{1}{q_r} - \frac{q_r}{(q_r + \eta)^2} \right]. \quad (101)$$

This is precisely condition (??), proving the stated necessary and sufficient condition. $\square$

Appendix B. Supplementary Material

SM.1.1 Determine the Number of Groups

We first use the observed user–product interaction network to recover latent groups of users and products. Importantly, the group-detection procedure is based only on whether a user has reviewed a product and does not use the numerical rating outcome. This separation ensures that the dependent variable in the subsequent empirical Bayes analysis does not mechanically determine the latent group memberships. Let

$$A \in \{0, 1\}^{N \times M}$$

denote the rectangular bi-adjacency matrix of the bipartite network, where N is the number of users and M is the number of products. Its (i, j) -th element is defined as

$$A_{ij} = \begin{cases} 1, & \text{if user } i \text{ rated product } j, \\ 0, & \text{otherwise.} \end{cases}$$

In the empirical application, the network consists of $N = 50,985$ users, $M = 25,848$ products, and $E = 412,947$ observed user–product links. The degree of user i and the degree of product j are respectively defined as

$$d_i = \sum_{j=1}^M A_{ij}, \quad q_j = \sum_{i=1}^N A_{ij}.$$

These degrees vary substantially across nodes, reflecting pronounced heterogeneity in user activity and product popularity.

To prevent such degree heterogeneity from being mistaken for genuine community structure, we employ a degree-corrected bipartite stochastic block model. The specification is motivated by the degree-corrected stochastic block model of ?, adapted here to a network with two distinct sets of nodes. Let

$$g_i \in \{1, \dots, K\}$$

denote the latent group membership of user i , and let

$$h_j \in \{1, \dots, L\}$$

denote the latent group membership of product j . In contrast to a one-mode block model, the numbers of groups on the two sides of the network are not required to be equal. Under the Poisson formulation of the degree-corrected model, the conditional edge intensity is

$$\mathbb{E}[A_{ij} \mid g_i, h_j] = \theta_i \phi_j \omega_{g_i h_j},$$

where θ_i captures the activity propensity of user i , ϕ_j captures the popularity propensity of product j , and ω_{kl} measures the underlying matching affinity between user group k and product group l . Thus, the expected interaction intensity can be interpreted as the product of user activity, product popularity, and group-level matching affinity.

The multiplicative representation requires a normalization because the degree parameters and block-affinity parameters are otherwise not separately identified. We impose the within-group normalizations

$$\sum_{i: g_i=k} \theta_i = 1, \quad k = 1, \dots, K,$$

and

$$\sum_{j: h_j=l} \phi_j = 1, \quad l = 1, \dots, L.$$

Define the total degree associated with user group k and product group l as

$$\kappa_k = \sum_{i: g_i=k} d_i, \quad \eta_l = \sum_{j: h_j=l} q_j,$$

and define the observed number of links between the corresponding pair of groups as

$$e_{kl} = \sum_{i: g_i=k} \sum_{j: h_j=l} A_{ij}.$$

Conditional on a given partition, the maximum-likelihood estimates of the node-specific degree parameters take the natural form

$$\hat{\theta}_i = \frac{d_i}{\kappa_{g_i}}, \quad \hat{\phi}_j = \frac{q_j}{\eta_{h_j}},$$

while, under the normalization above, the block-intensity estimator is proportional to the corresponding block-edge count e_{kl} . Consequently, heterogeneity in the overall activity of individual users and the popularity of individual products is absorbed by the node-specific parameters

rather than by the latent group assignments.

Profiling out the degree parameters yields a convenient objective for evaluating a candidate partition. Let

$$E = \sum_{i=1}^N \sum_{j=1}^M A_{ij}$$

denote the total number of observed edges. Conditional on the degree sequence, the expected number of edges between user group k and product group l under random bipartite matching is

$$e_{kl}^{(0)} = \frac{\kappa_k \eta_l}{E}.$$

We therefore evaluate a candidate partition using the degree-corrected profile objective

$$Q_{\text{DC}}(K, L) = \sum_{k=1}^K \sum_{l=1}^L e_{kl} \log \left(\frac{e_{kl}}{e_{kl}^{(0)}} \right) = \sum_{k=1}^K \sum_{l=1}^L e_{kl} \log \left(\frac{e_{kl} E}{\kappa_k \eta_l} \right),$$

where a term with $e_{kl} = 0$ is defined to be zero. The factor E inside the logarithm contributes only the constant $E \log E$ across candidate partitions and therefore does not affect the ranking of alternative group assignments. Economically, Q_{DC} measures the extent to which observed user-group–product-group interactions differ from what would be expected solely from heterogeneous user activity and product popularity.

Direct maximization of the block-model likelihood over all possible user and product assignments is computationally infeasible for a network of this size. We therefore estimate the latent memberships using degree-normalized spectral co-clustering, following the normalized asymmetric clustering framework of [Rohe et al. \(2016\)](#). Let

$$D_u = \text{diag}(d_1, \dots, d_N), \quad D_p = \text{diag}(q_1, \dots, q_M),$$

and construct the normalized bi-adjacency matrix

$$\tilde{A} = D_u^{-1/2} A D_p^{-1/2}.$$

We then compute its truncated singular value decomposition,

$$\tilde{A} = U \Sigma V^\top.$$

The left singular vectors contain the spectral information associated with users, whereas the right singular vectors contain the corresponding information for products. For the normalized network, the leading singular direction primarily represents the global degree component and is therefore excluded from the clustering step.

For each candidate pair (K, L) , we define

$$r = \min\{K, L\},$$

and construct the nontrivial user and product embeddings as

$$U^* = [u_2, \dots, u_r], \quad V^* = [v_2, \dots, v_r].$$

The rows of U^* and V^* are normalized to unit Euclidean length before clustering. We then cluster the rows of U^* into K user groups and independently cluster the rows of V^* into L product groups. In the implementation, K -means-type clustering is performed separately on the two embeddings, with MiniBatch K -means used for computational scalability. Because the K -means objective can be sensitive to initialization, several independent initializations are considered for each candidate pair (K, L) . For a fixed (K, L) , we retain the user and product assignments that produce the largest value of $Q_{\text{DC}}(K, L)$.

Although the degree-corrected profile objective increases the fit of the model, it cannot by itself determine the appropriate numbers of groups, because larger values of K and L generally provide greater flexibility. We therefore introduce a complexity penalty motivated by the corrected Bayesian information criterion of ?. Their one-mode CBIC explicitly penalizes the latent community assignments in addition to the usual model-parameter complexity. In a one-mode network with n nodes and k groups, there are approximately k^n possible membership configurations, whose logarithm is $n \log k$. In the present bipartite setting, the user side has approximately K^N possible assignments and the product side has approximately L^M possible assignments. Hence, their combined log assignment complexity is

$$N \log K + M \log L.$$

Because our network is rectangular, the block-affinity matrix contains KL parameters rather than the $k(k+1)/2$ parameters arising from a symmetric one-mode block matrix. Motivated by

this distinction, we use the following degree-corrected bipartite adaptation of the CBIC:

$$\text{CBIC}_{\text{DC}}(K, L) = -2Q_{\text{DC}}(K, L) + 2N \log K + 2M \log L + KL \log(NM).$$

The first term rewards goodness of fit, the second and third terms penalize the uncertainty associated with user and product membership assignments, respectively, and the final term penalizes the KL block-affinity parameters. We emphasize that this exact rectangular expression is a bipartite adaptation of the CBIC principle rather than a formula stated verbatim in [Hu et al. \(2020\)](#). The numbers of user and product groups are therefore selected jointly according to

$$(\widehat{K}, \widehat{L}) = \arg \min_{K, L} \text{CBIC}_{\text{DC}}(K, L).$$

In the empirical implementation, we first examine a broad set of candidate values for K and L , and subsequently conduct a finer search around the region attaining the smallest criterion values. Multiple clustering initializations are evaluated for each candidate specification, and the partition attaining the highest degree-corrected profile objective is used to calculate the corresponding CBIC. This procedure selects

$$\widehat{K} = 10, \quad \widehat{L} = 11,$$

indicating that the latent structure of the user side and product side cannot be adequately represented by imposing an equal number of groups. Because cluster labels are arbitrary up to permutation, the estimated block structure is not required to be diagonal: for example, user group k may have its strongest interaction affinity with product group $l \neq k$. Accordingly, the economically relevant objects are the estimated group-to-group affinities rather than the numerical labels assigned to the groups.

Finally, the estimated memberships are used as inputs to the empirical Bayes estimation of heterogeneous user and product effects. Let

$$\widehat{Z}_u \in \{0, 1\}^{N \times \widehat{K}}$$

and

$$\widehat{Z}_p \in \{0, 1\}^{M \times \widehat{L}}$$

denote the estimated one-hot membership matrices for users and products, respectively. We combine them into

$$\hat{Z} = \begin{pmatrix} \hat{Z}_u & 0 \\ 0 & \hat{Z}_p \end{pmatrix} \in \{0, 1\}^{(N+M) \times (\hat{K} + \hat{L})}.$$

Since $\hat{K} = 10$ and $\hat{L} = 11$, the subsequent empirical Bayes analysis contains 21 group-specific distributions of heterogeneous effects. The estimated network memberships therefore determine which users and products are pooled together when learning the group-specific prior distributions, while the observed rating equation is subsequently used to estimate the magnitude of individual user and product effects.

SM.2.2 Estimated Group Structure in the Amazon User–Product Network

We next examine the latent group structure of the Amazon user–product interaction network. Let $B \in \{0, 1\}^{N \times M}$ denote the bipartite incidence matrix, where $B_{ij} = 1$ if user i rates product j , and $B_{ij} = 0$ otherwise. Group detection is based exclusively on the topology of this interaction network and therefore does not use the numerical rating outcomes. This separation allows the estimated network memberships to be subsequently used in the empirical-Bayes estimation without mechanically constructing the groups from the dependent variable.

To account for the substantial heterogeneity in user activity and product popularity, we apply the degree-corrected bipartite spectral clustering procedure described in Section 3.1. In particular, we construct the degree-normalized biadjacency matrix

$$\tilde{B} = D_u^{-1/2} B D_p^{-1/2},$$

compute its leading singular vectors, and cluster users and products separately using their corresponding spectral representations. The numbers of groups on the two sides are allowed to differ. Comparing alternative candidate specifications using the degree-corrected model-selection criterion yields

$$\hat{K} = 10, \quad \hat{L} = 11,$$

corresponding to ten user groups and eleven product groups.

Figure 4 displays the incidence matrix after reordering users and products according to their estimated group memberships. The visible concentration of links within particular user-group–product-group combinations provides direct evidence of non-uniform matching patterns across the network. Importantly, because the labels of the estimated groups are arbitrary, the strongest interaction blocks need not lie on the main diagonal. For example, user group $U1$ interacts particularly intensively with product group $P4$, while other prominent combinations include $U8$ – $P6$, $U3$ – $P5$, $U7$ – $P8$, and $U10$ – $P10$.

To quantify these differences, Figure 5 reports the density of each estimated user-group–product-group block. For a user group k and product group l , the block density is defined as

$$\hat{\rho}_{kl} = \frac{e_{kl}}{N_k M_l},$$

where e_{kl} denotes the number of observed links between the two groups, and N_k and M_l de-

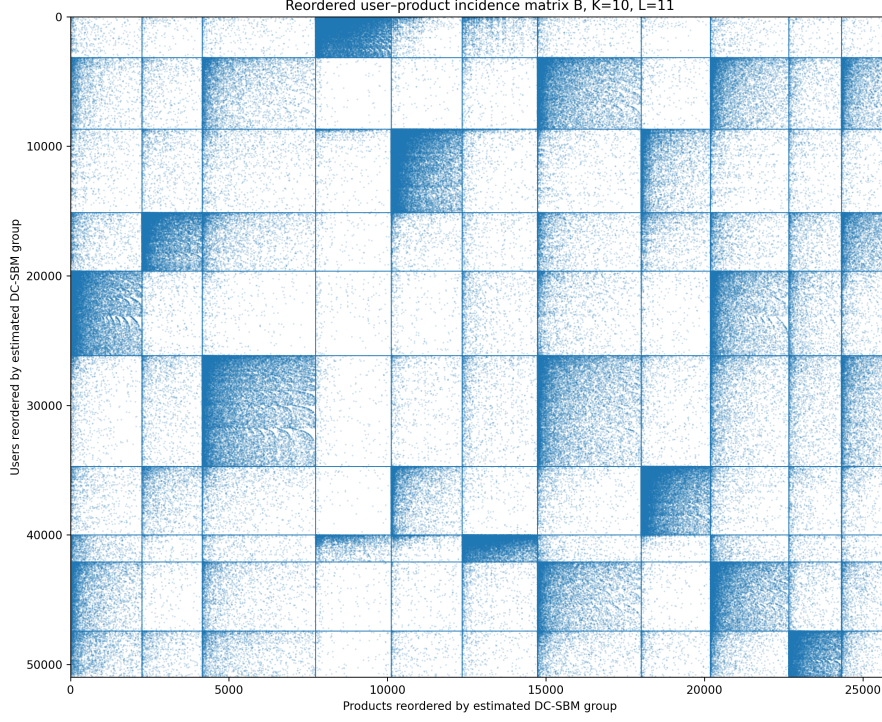


Figure 4: Reordered Amazon user-product incidence matrix. Users and products are reordered according to the estimated degree-corrected bipartite group memberships, with $\widehat{K} = 10$ user groups and $\widehat{L} = 11$ product groups.

note the numbers of users and products in the corresponding groups. The heatmap reveals pronounced variation in interaction intensities across blocks. The largest estimated density occurs between $U1$ and $P4$, with a density of approximately 0.003689, followed by $U8-P6$ with 0.003374, $U3-P5$ with 0.002531, and $U7-P8$ with 0.002527. These patterns suggest that the network contains systematic group-level matching structure beyond differences in individual node degrees.

Table 3 reports the corresponding numbers of observed links in each block. The edge counts display substantial heterogeneity across group pairs. For instance, the $U3-P5$ block contains 36,466 observed interactions, the $U7-P8$ block contains 29,279, and the $U1-P4$ block contains 27,755. The block counts should be interpreted jointly with the density measures, since raw interaction counts also depend on the sizes of the corresponding user and product groups.

Table 4 reports the corresponding block densities. Unlike the raw counts, these statistics normalize for the number of potential user-product links within each group pair and therefore provide a more direct measure of the strength of group-level matching.

Overall, the reordered incidence matrix and block-density estimates provide consistent evidence of pronounced group-level interaction patterns in the Amazon user-product network. The

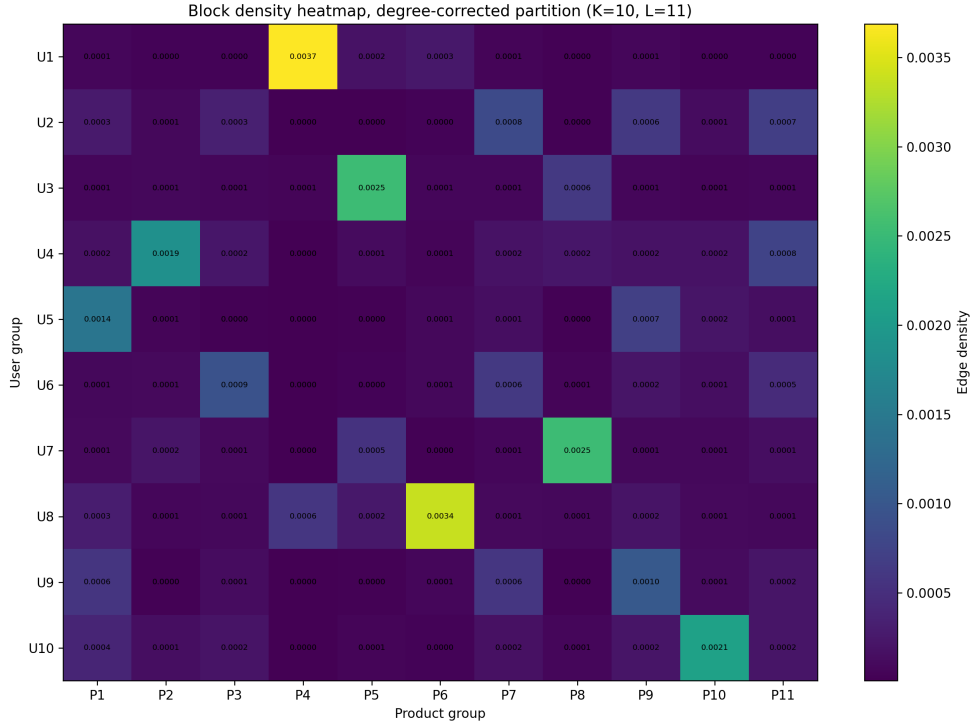


Figure 5: Block-density heatmap for the estimated degree-corrected bipartite partition with $\hat{K} = 10$ user groups and $\hat{L} = 11$ product groups. Each cell reports the observed edge density between the corresponding user and product groups.

Table 3: Observed Edge Counts by Estimated User and Product Groups

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11
U1	498	215	483	27,755	1,328	1,872	645	183	524	207	192
U2	3,378	879	6,443	136	222	480	15,058	381	8,415	1,213	5,849
U3	1,096	1,022	1,718	979	36,466	1,144	1,524	8,685	1,211	684	738
U4	1,803	15,990	3,647	199	1,215	562	2,697	2,251	2,019	1,370	5,192
U5	21,197	694	588	190	354	1,252	2,922	403	11,442	2,351	1,412
U6	1,539	1,594	28,651	406	939	1,630	17,189	1,107	4,502	2,114	5,995
U7	1,274	2,243	1,863	88	6,507	400	1,325	29,279	1,199	798	1,209
U8	1,433	269	671	2,962	1,159	16,710	720	402	1,013	356	294
U9	6,872	316	2,417	181	370	937	10,230	333	13,603	1,004	1,692
U10	3,093	982	2,493	112	451	361	2,066	607	1,923	12,532	1,159

Notes: Entries report the number of observed user-product links between estimated user group Uk and product group Pl .

Table 4: Edge Densities by Estimated User and Product Groups

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11
U1	0.000070	0.000036	0.000043	0.003689	0.000188	0.000250	0.000063	0.000027	0.000067	0.000040	0.000040
U2	0.000271	0.000084	0.000326	0.000010	0.000018	0.000037	0.000833	0.000031	0.000615	0.000133	0.000692
U3	0.000076	0.000084	0.000075	0.000064	0.002531	0.000075	0.000072	0.000617	0.000076	0.000064	0.000075
U4	0.000177	0.001861	0.000226	0.000018	0.000120	0.000052	0.000183	0.000228	0.000181	0.000183	0.000753
U5	0.001443	0.000056	0.000025	0.000012	0.000024	0.000081	0.000137	0.000028	0.000709	0.000218	0.000142
U6	0.000080	0.000098	0.000936	0.000020	0.000049	0.000080	0.000614	0.000059	0.000213	0.000149	0.000458
U7	0.000107	0.000223	0.000098	0.000007	0.000548	0.000032	0.000077	0.002527	0.000092	0.000091	0.000149
U8	0.000305	0.000068	0.000090	0.000595	0.000248	0.003374	0.000106	0.000088	0.000196	0.000103	0.000092
U9	0.000574	0.000031	0.000127	0.000014	0.000031	0.000074	0.000589	0.000029	0.001035	0.000114	0.000208
U10	0.000384	0.000144	0.000195	0.000013	0.000056	0.000042	0.000176	0.000077	0.000217	0.002115	0.000212

Notes: For user group k and product group l , the reported quantity is $\hat{\rho}_{kl} = e_{kl}/(N_k M_l)$, where e_{kl} is the observed block-edge count and $N_k M_l$ is the number of possible links within the corresponding block.

estimated structure is neither purely diagonal nor driven by a single high-density block; instead, different user groups display distinct affinities toward different product groups. This finding supports the use of separate latent groups on the user and product sides and provides the first-stage network classification used in the subsequent empirical-Bayes analysis. In particular, the estimated group memberships determine the group-specific prior distributions toward which the individual user and product effects are selectively shrunk.

SM.3.3 Group-Specific Empirical-Bayes Estimation of User and Product Effects

Building on the network partition described in the preceding subsection, we estimate group-specific distributions of user and product heterogeneity in the Amazon Industrial and Scientific ratings sample. We condition on the supplied partition into $K = 10$ user groups and $L = 11$ product groups throughout this stage. The estimation sample contains $R = 412,947$ rating observations from $N = 50,985$ users on $M = 25,848$ products, identified by `parent_asin`. The mean observed rating is 4.450578. The estimated memberships provide group-specific reference distributions for regularizing individual effects while allowing their locations and dispersions to differ across groups.

For observed rating r , let $i(r)$ and $j(r)$ identify its user and product. We specify the additive model

$$y_r = \beta_0 + \alpha_{i(r)} + \gamma_{j(r)} + \varepsilon_r, \quad r = 1, \dots, R, \quad (102)$$

where α_i and γ_j capture user and product heterogeneity, respectively. The Gaussian working likelihood assumes independent errors with mean zero and common variance σ^2 , independent of the effects conditional on the network and group memberships. Conditional on the estimated memberships $\hat{g}_i$ and $\hat{h}_j$, the effects have independent group-specific priors

$$\alpha_i \mid \hat{g}_i = k \sim \mathcal{N}(\mu_k^s, (\tau_k^s)^2), \quad \gamma_j \mid \hat{h}_j = \ell \sim \mathcal{N}(\mu_\ell^t, (\tau_\ell^t)^2). \quad (103)$$

The model therefore contains 21 group-specific normal distributions. Group labels retain the supplied numbering and have no intrinsic ranking.

We use the preliminary unrestricted two-way fixed-effects (FE) estimates to obtain the intercept and innovation variance. The FE stage imposes the unweighted normalizations $\sum_i \hat{\alpha}_i^{\text{FE}} = 0$ and $\sum_j \hat{\gamma}_j^{\text{FE}} = 0$ and yields $\hat{\beta}_0^{\text{FE}} = 4.425580$. The observed bipartite graph is connected, so the residual degrees of freedom are $R - (N + M - 1) = 336,115$. The corresponding variance estimate is

$$\hat{\sigma}^2 = \frac{\sum_{r=1}^R (y_r - \hat{\beta}_0^{\text{FE}} - \hat{\alpha}_{i(r)}^{\text{FE}} - \hat{\gamma}_{j(r)}^{\text{FE}})^2}{R - (N + M - 1)} = 0.846414. \quad (104)$$

Both quantities are held fixed throughout the subsequent estimation. Displayed values are rounded; the computation uses their stored full precision.

We then implement the plug-in version of the marginal-likelihood method in Section 3.2. Let $\tilde{Y} = Y - \hat{\beta}_0^{\text{FE}} \mathbf{1}_R$, and let B be the $R \times (N + M)$ rating-level incidence matrix, with one user

entry and one product entry equal to one in each row. Stack the effects as $\theta = (\alpha', \gamma')'$, and let $\hat{Z}$ map nodes to their estimated groups. Writing the group means and variances as μ_g and τ_g^2 , define $D = \text{diag}(\hat{Z}\tau_g^2)$. Integrating out the node effects gives the working marginal model

$$\tilde{Y} \mid B, \hat{Z} \sim \mathcal{N}(B\hat{Z}\mu_g, \Omega), \quad \Omega = BDB' + \hat{\sigma}^2 I_R. \quad (105)$$

The group means require a location normalization even though the intercept is fixed: adding a common constant to all user-group means and subtracting it from all product-group means leaves $B\hat{Z}\mu_g$ unchanged. Following the paper, we impose

$$\sum_{\ell=1}^L \frac{M_\ell}{M} \mu_\ell^t = 0, \quad M_\ell = \sum_{j=1}^M \mathbf{1}\{\hat{h}_j = \ell\}. \quad (106)$$

Thus, the 21 reported group means contain 20 free location parameters. No additional mean-zero restriction is imposed on the user-group means.

The group parameters are estimated jointly by maximizing

$$\begin{aligned} \ell(\mu_g, \tau_g^2) = & -\frac{R}{2} \log(2\pi) - \frac{1}{2} \log |\Omega| \\ & - \frac{1}{2} (\tilde{Y} - B\hat{Z}\mu_g)' \Omega^{-1} (\tilde{Y} - B\hat{Z}\mu_g), \end{aligned} \quad (107)$$

subject to equation (106) and nonnegative group variances. For each variance vector, the group means are obtained by constrained generalized least squares. The resulting profile likelihood is optimized over the 21 group variances. This is marginal maximum likelihood with fixed plug-in nuisance parameters, rather than restricted maximum likelihood or joint estimation of the innovation variance.

The criterion uses the original ratings and the full covariance induced by shared users and products. It therefore incorporates the network dependence that is omitted when preliminary FE estimates are treated as independent signals with sampling-variance proxies $\hat{\sigma}^2/d_i$. Computation uses exact determinant and Woodbury identities, followed by elimination of the diagonal user block and Cholesky factorization of the product-side Schur complement. This avoids forming the $R \times R$ observation covariance matrix. Analytic variance scores are evaluated using exact inverse-diagonal terms.

Model-level results. Table 5 summarizes the sample, fixed inputs, and convergence diagnostics. The marginal-likelihood optimization converges in eight iterations using nine likelihood evaluations. The largest absolute score with respect to the log group variances is 8.995×10^{-6} , and all estimated variances are strictly positive. The likelihood and derivative calculations were checked against direct covariance-matrix calculations on a small test model. These checks and the reported scores establish numerical consistency and convergence of the implementation; they do not constitute a general proof of global optimality for variance-component likelihoods.

Table 5: Sample characteristics, fixed inputs, and marginal-likelihood diagnostics.

Statistic	Value
Rating observations	412,947
Users	50,985
Products	25,848
User groups	10
Product groups	11
Connected components	1
Mean observed rating	4.450578
Fixed FE intercept	4.425580
Fixed innovation variance (df corrected)	0.846414
FE residual degrees of freedom	336,115
Free group-mean parameters	20
Group-variance parameters	21
Marginal log likelihood	-586,357.855799
Optimization iterations	8
Likelihood evaluations	9
Largest absolute log-variance score	8.995×10^{-6}
Largest absolute constrained mean score	1.689×10^{-12}

Notes: The intercept and innovation variance are fixed at their preliminary FE estimates. The marginal log likelihood includes the Gaussian constant. Scores are derivatives of the likelihood criterion, rather than changes in individual effects between iterations. Values are rounded for presentation.

Estimated group-specific heterogeneity. Tables 6 and 7 report the estimated group-specific prior distributions. User prior means range from -0.170941 in U3 to 0.144425 in U1, a difference of 0.315367 rating points. Holding product heterogeneity constant, these estimates indicate systematic differences in users' rating tendencies. The network partition therefore contains information about rating behavior even though the groups are estimated from the interaction structure rather than from rating outcomes.

Estimated within-group user variances also differ substantially. U1 has the smallest variance, 0.024957 , corresponding to a prior standard deviation of 0.157978 , whereas U3 has the largest variance, 0.478668 , corresponding to a standard deviation of 0.691859 . U7 also exhibits relatively high dispersion, with prior variance 0.351302 . Thus, a relatively favorable group-level

rating tendency can coexist with comparatively homogeneous user effects, while a group with a negative mean can exhibit considerable individual variation. Allowing both prior locations and dispersions to vary makes the targets and strength of regularization group specific.

Table 6: Estimated user-group prior parameters from plug-in marginal maximum likelihood.

Group	Nodes	$\hat{\mu}$	$\hat{\tau}^2$	$\hat{\tau}$
U1	3,152	0.144425	0.024957	0.157978
U2	5,531	0.060736	0.165517	0.406838
U3	6,429	-0.170941	0.478668	0.691859
U4	4,515	0.035314	0.205083	0.452861
U5	6,525	0.015160	0.242132	0.492069
U6	8,558	0.062127	0.171099	0.413641
U7	5,295	-0.074558	0.351302	0.592708
U8	2,085	0.099826	0.088797	0.297988
U9	5,314	0.032102	0.203814	0.451458
U10	3,581	0.011992	0.225873	0.475261

Notes: The columns report group sizes and estimated prior means, variances, and standard deviations. Prior SD measures within-group heterogeneity, not the standard error of the estimated prior mean. Means follow the normalization in equation (106). Values are rounded to six decimal places.

Product prior means range from -0.178940 in P8 to 0.120663 in P4, a difference of 0.299603 rating points. P5 also has a low prior mean, -0.175566 . Conditional on user-specific rating tendencies, products in P4 therefore have a higher average product-specific rating component than products in P5 and P8. The product groups inferred from the network are informative about systematic variation in product evaluations.

Product prior variances show substantial differences in within-group dispersion. P4 has the smallest variance, 0.008926 , corresponding to a prior standard deviation of 0.094478 . P8 and P5 have the largest variances, 0.149291 and 0.145862 , with standard deviations of 0.386382 and 0.381919 , respectively. Product effects in P4 are therefore much more concentrated around their group mean. The larger variances in P5 and P8 allow substantial product-level heterogeneity within these groups. These effects describe conditional rating components and should not be interpreted as causal measures of product quality or user behavior.

Comparison with the FE-based approximation. To compare prior means across implementations, we first express both sets of estimates under the normalization in equation (106). This requires adding 0.006094 to the previously reported user-group means and subtracting the same constant from the product-group means; the full-precision adjustment leaves all user-product mean sums unchanged. After alignment, the largest absolute difference between corresponding group means is 0.019819 rating points. The estimated prior locations are therefore numerically

Table 7: Estimated product-group prior parameters from plug-in marginal maximum likelihood.

Group	Nodes	$\hat{\mu}$	$\hat{\tau}^2$	$\hat{\tau}$
P1	2,252	-0.021073	0.070295	0.265132
P2	1,903	0.006960	0.059220	0.243352
P3	3,576	0.064244	0.047835	0.218712
P4	2,387	0.120663	0.008926	0.094478
P5	2,241	-0.175566	0.145862	0.381919
P6	2,375	-0.011936	0.050207	0.224070
P7	3,270	0.073730	0.039899	0.199747
P8	2,188	-0.178940	0.149291	0.386382
P9	2,473	0.008051	0.063521	0.252033
P10	1,655	0.016329	0.068365	0.261468
P11	1,528	0.027312	0.054498	0.233448

Notes: The columns report group sizes and estimated prior means, variances, and standard deviations. Prior SD measures within-group heterogeneity, not the standard error of the estimated prior mean. Means follow the normalization in equation (106). Values are rounded to six decimal places.

close across the two methods.

The variance estimates differ more noticeably. All 21 variances are lower under the full marginal likelihood. User-group variances decline by 1.33%–7.63%, with a median reduction of 3.11%. Product-group variances decline by 5.65%–40.01%, with a median reduction of 13.51%. The largest proportional change occurs in P4, whose estimated variance decreases from 0.014880 to 0.008926. The approximation thus delivers similar prior locations in this sample, but it is not numerically equivalent to the full marginal-likelihood method, particularly for product-side dispersion.

Empirical-Bayes shrinkage under the fitted priors. Given the estimated group parameters, let $\hat{D} = \text{diag}(\hat{Z}\hat{\tau}_g^2)$. The associated empirical-Bayes posterior mean is

$$\hat{\theta}^{\text{EB}} = \hat{Z}\hat{\mu}_g + \hat{D}B'(B\hat{D}B' + \hat{\sigma}^2 I_R)^{-1}(\tilde{Y} - B\hat{Z}\hat{\mu}_g). \quad (108)$$

The intercept remains fixed at $\hat{\beta}_0^{\text{FE}}$. Equivalently, the posterior mean minimizes

$$Q(\alpha, \gamma) = \frac{1}{2\hat{\sigma}^2} \sum_{r=1}^R (y_r - \hat{\beta}_0^{\text{FE}} - \alpha_{i(r)} - \gamma_{j(r)})^2 + \frac{1}{2} \sum_{i=1}^N \frac{(\alpha_i - \hat{\mu}_{g_i}^s)^2}{(\hat{\tau}_{g_i}^s)^2} + \frac{1}{2} \sum_{j=1}^M \frac{(\gamma_j - \hat{\mu}_{h_j}^t)^2}{(\hat{\tau}_{h_j}^t)^2}. \quad (109)$$

The Gaussian posterior mean and mode coincide conditional on these plug-in inputs.

The conditional update for user i can be written as

$$\begin{aligned}\hat{\alpha}_i^{\text{EB}} &= w_i \left[\frac{1}{d_i} \sum_{r:i(r)=i} \left(y_r - \hat{\beta}_0^{\text{FE}} - \hat{\gamma}_{j(r)}^{\text{EB}} \right) \right] + (1 - w_i) \hat{\mu}_{g_i}^s, \\ w_i &= \frac{(\hat{\tau}_{g_i}^s)^2}{(\hat{\tau}_{g_i}^s)^2 + \hat{\sigma}^2/d_i},\end{aligned}\tag{110}$$

where d_i is the user's number of ratings. An analogous expression applies to products. For a fixed rating count, a smaller prior variance places greater weight on the group mean. For a fixed prior variance, more ratings place greater weight on the individual's adjusted outcomes. Because those outcomes depend on the jointly determined effects on the opposite side, the posterior estimator generally cannot be expressed as a scalar weighted average of a preliminary FE estimate and its group mean. The scalar weights characterize conditional updates; they are not substituted for the network covariance in hyperparameter estimation.

Uncertainty and interpretation. Conditional on the fixed intercept, innovation variance, memberships, and estimated hyperparameters, the full Gaussian posterior covariance is

$$\hat{V} = \left(\frac{B'B}{\hat{\sigma}^2} + \hat{D}^{-1} \right)^{-1}.\tag{111}$$

Its diagonal elements account for posterior dependence propagated through the network. In contrast, the scalar expression $(d_v/\hat{\sigma}^2 + 1/\hat{\tau}_{q(v)}^2)^{-1}$ is the conditional variance of node v when the other node effects are held fixed. Neither measure incorporates estimation uncertainty in the intercept, innovation variance, memberships, or prior parameters.

Overall, the fitted marginal model reveals substantial heterogeneity in both the locations and dispersions of the group-specific priors. The resulting shrinkage targets differ across network-defined groups, and the estimated dispersion parameters allow the strength of pooling to vary across groups and between users and products. These findings characterize the fitted distribution of heterogeneity. Assessing gains in predictive performance or effect-estimation accuracy requires additional evaluation beyond the reported marginal-likelihood results.